

Beelink GTR9 Pro

UEFI BIOS Setup Reference

Complete catalog of every Setup option exposed by the GTR9 Pro (v2.2) firmware

Document 1.6.5 · AMD Strix Halo · Ryzen™ AI Max+ 395 · AMI Aptio V

Board	Beelink GTR9 Pro v2.2 (Realtek RTL8127 NIC)
BIOS image of record	GTRPRPI1001C (32 MB); prior image GTRPR07
BIOS vendor	American Megatrends International — AMI Aptio V
SoC	AMD Strix Halo — Ryzen AI Max+ 395
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How to Read This Catalog

Structure

Chapter	One firmware form-set
Sub-section	One form — a single Setup page, in firmware presentation order
Information fields	Read-only values and links to child pages, summarised above each table
Settings table	Configurable options: Setting / Type / Options-Values-Range

Type column

Selection	One-of list
Numeric	Value within a Min / Max / Step range
Checkbox	Boolean toggle
Text / Password	Free-form string entry
Ordered List	Reorderable priority list
Date / Time	Calendar or clock field

Values

Each selection choice prints its stored NVRAM value in [brackets]; numeric settings print Min / Max / Step with the compiled default. Bracketed values are written directly to NVRAM and are usable for AMISCE / SCEWIN scripting.

Markers

■ default	Factory compiled Standard default value, on the default option
■ performance	Recommended value for the Ryzen AI Max+ 395, with a one-line rationale
CHANGE	Change away from default for a clear gain
TUNE	Performance-relevant but workload-specific or expert-only; record originals first
KEEP	Default already favours performance; leave it

Notes

- Many options are conditionally hidden (suppress-if / grayout-if); this catalog is a superset of what any single unit displays.
- Defaults shown are compiled Standard Defaults; a unit's live values may differ.
- AMD CBS, AMD Overclocking and AMD PMF expose low-level controls — change only what you understand.

Aptio Setup (Main Menu Tree)

35 forms · 161 settings

Setup (0x2710)

Submenus Main Advanced Chipset Security Boot Save & Exit

Main (0x2711)

Information fields BIOS Vendor, Core Version, Compliancy, BIOS Version, Build Date and Time, Access Level, Serial Number, Total Memory

Setting	Type	Options / Values / Range
System Language	Selection	(populated at runtime) [0x0] (populated at runtime) [0x1]
System Date	Date	Date field (MM/DD/YYYY)
System Time	Time	Time field (HH:MM:SS)

Advanced (0x2712)

Submenus Trusted Computing ■ Trusted Computing ■ Trusted Computing ■ AMD fTPM configuration ■ ACPI Settings ■ AMD CBS ■ AMD PBS ■ AMD PMF Feature ■ Dash Function ■ Super IO Configuration ■ Hardware Monitor ■ Watch Dog Configuration ■ S5 RTC Wake Settings ■ UEFI Variables Protection ■ CPU Configuration ■ AmiAgesaChipsetWrapper Configuration ■ AMI Graphic Output Protocol Policy ■ Debug Port Table Configuration ■ SIO Common Setting ■ PCI Subsystem Settings ■ USB Configuration ■ Network Stack Configuration ■ NVMe Configuration ■ Demo Board

Trusted Computing (0x2719)

Information fields NO Security Device Found

Information fields Firmware Version:, Vendor:, TPM Enabled Status:, TPM Active Status:, TPM Owner Status:, TCM Enabled Status:, TCM Active Status:, TCM Owner Status:

Information fields Firmware Version:, Vendor:, Active PCR banks, Available PCR banks

Setting	Type	Options / Values / Range
Security Device Support	Selection	Disabled [0x0] Enabled [0x1] ■ default

Trusted Computing (0x271A)

Information fields NO Security Device Found

Information fields Firmware Version:, Vendor:, TPM Enabled Status:, TPM Active Status:, TPM Owner Status:, TCM Enabled Status:, TCM Active Status:, TCM Owner Status:

Information fields Firmware Version:, Vendor:, Active PCR banks, Available PCR banks

Setting	Type	Options / Values / Range
Security Device Support	Selection	Disabled [0x0] Enabled [0x1] ■ default
TPM State	Selection	Disabled [0x0] Enabled [0x1] ■ default
Pending operation	Selection	None [0x0] ■ default TPM Clear [0x5]
TCM State	Selection	Disabled [0x0] Enabled [0x1] ■ default
Device Select	Selection	TPM 1.2 [0x0] TPM 2.0 [0x1] Auto [0x2] ■ default

Trusted Computing (0x271B)

Information fields NO Security Device Found

Information fields Firmware Version:, Vendor:, TPM Enabled Status:, TPM Active Status:, TPM Owner Status:, TCM Enabled Status:, TCM Active Status:, TCM Owner Status:

Information fields Firmware Version:, Vendor:, Active PCR banks, Available PCR banks

Setting	Type	Options / Values / Range
Security Device Support	Selection	Disabled [0x0] Enabled [0x1] ■ default
SHA-1 PCR Bank	Selection	Disabled [0x0] ■ default Enabled [0x1]
SHA256 PCR Bank	Selection	Disabled [0x0] Enabled [0x2] ■ default
SHA384 PCR Bank	Selection	Disabled [0x0] ■ default Enabled [0x4]
SHA512 PCR Bank	Selection	Disabled [0x0] ■ default Enabled [0x8]
SM3_256 PCR Bank	Selection	Disabled [0x0] ■ default Enabled [0x10]
Pending operation	Selection	None [0x0] ■ default TPM Clear [0x1]
Platform Hierarchy	Selection	Disabled [0x0] Enabled [0x1] ■ default
Storage Hierarchy	Selection	Disabled [0x0] Enabled [0x1] ■ default
Endorsement Hierarchy	Selection	Disabled [0x0] Enabled [0x1] ■ default
Physical Presence Spec Version	Selection	1.2 [0x0] 1.3 [0x1] ■ default
TPM 2.0 UEFI Spec Version	Selection	TCG_1_2 [0x1] TCG_2 [0x2] ■ default
TPM 2.0 InterfaceType	Selection	CRB [0x0] ■ default TIS [0x1]
Device Select	Selection	TPM 1.2 [0x0] TPM 2.0 [0x1] Auto [0x2] ■ default

AMD fTPM configuration (0x271C)

Setting	Type	Options / Values / Range
Erase fTPM NV for factory reset	Selection	Disabled [0x0] Enabled [0x1] ■ default

ACPI Settings (0x271D)

Setting	Type	Options / Values / Range
ACPI Sleep State	Selection	Suspend Disabled [0x0] S3 (Suspend to RAM) [0x2] ■ default

Hardware Monitor (0x2735)

Submenus Smart Fan Function

Information fields CPU temperature, System temperature, CPU Fan1 Speed, CPU Fan2 Speed, VCore, V_SM, 3.3V, 5V

Smart Fan Function (0x2737)

Setting	Type	Options / Values / Range
CPU Smart Fan1 Mode	Selection	Manual Mode [0x0] Automatic Mode [0x1] ■ default Full on Mode [0x2] ■ performance TUNE Automatic (curve) for balance; Full-On maximizes cooling/boost headroom at the cost of noise. Manual lets you set an aggressive fixed curve.
Manual PWM Setting	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x7F ■ default ■ performance TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temp / higher PWM) to gain thermal headroom; trade against noise.
Fan off temperature limit	Numeric	Min 0x0 · Max 0x7F · Step 0x1 · Default 0x1E ■ default ■ performance TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temp / higher PWM) to gain thermal headroom; trade against noise.
Fan start temperature limit	Numeric	Min 0x0 · Max 0x7F · Step 0x1 · Default 0x23 ■ default ■ performance TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temp / higher PWM) to gain thermal headroom; trade against noise.
Fan full speed temperature limit	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x5B ■ default ■ performance TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temp / higher PWM) to gain thermal headroom; trade against noise.
Fan start PWM	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x46 ■ default ■ performance TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temp / higher PWM) to gain thermal headroom; trade against noise.
PWM SLOPE SETTING	Selection	0.125 PWM [0x0] 0.25 PWM [0x1] 0.5 PWM [0x2] 1 PWM [0x3] ■ default 2 PWM [0x4] 4 PWM [0x5] 8 PWM [0x6] 15.875 PWM [0x7] 4.5 PWM [0x8] ■ performance TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temp / higher PWM) to gain thermal headroom; trade against noise.
Fan start PWM	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x64 ■ default ■ performance TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temp / higher PWM) to gain thermal headroom; trade against noise.

Setting	Type	Options / Values / Range
CPU Smart Fan2 Mode	Selection	Manual Mode [0x0] Automatic Mode [0x1] ■ default Full on Mode [0x2] ■ performance TUNE Automatic for balance, Full-On for maximum thermal headroom (louder). Match to Fan1.
PWM SLOPE SETTING	Selection	0.125 PWM [0x0] 0.25 PWM [0x1] 0.5 PWM [0x2] 1 PWM [0x3] ■ default 2 PWM [0x4] 4 PWM [0x5] 8 PWM [0x6] 15.875 PWM [0x7] ■ performance TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temp / higher PWM) to gain thermal headroom; trade against noise.
Smart Fan 3 Mode	Selection	Manual Mode [0x0] ■ default Automatic Mode [0x1] ■ performance TUNE Automatic for balance; aggressive/Full-On if this fan cools a perf-critical zone.
Fan off temperature limit	Numeric	Min 0x0 · Max 0x7F · Step 0x1 · Default 0x10 ■ default ■ performance TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temp / higher PWM) to gain thermal headroom; trade against noise.
Fan start temperature limit	Numeric	Min 0x0 · Max 0x7F · Step 0x1 · Default 0x20 ■ default ■ performance TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temp / higher PWM) to gain thermal headroom; trade against noise.
Fan full speed temperature limit	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x7F ■ default ■ performance TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temp / higher PWM) to gain thermal headroom; trade against noise.
Fan start PWM	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x7F ■ default ■ performance TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temp / higher PWM) to gain thermal headroom; trade against noise.
PWM SLOPE SETTING	Selection	0.125 PWM [0x0] 0.25 PWM [0x1] 0.5 PWM [0x2] ■ default 1 PWM [0x3] 2 PWM [0x4] 4 PWM [0x5] 8 PWM [0x6] 15.875 PWM [0x7] ■ performance TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temp / higher PWM) to gain thermal headroom; trade against noise.
Smart Fan 4 Mode	Selection	Manual Mode [0x0] ■ default Automatic Mode [0x1] ■ performance TUNE Automatic for balance; aggressive curve to hold temps down under sustained load.

Watch Dog Configuration (0x2738)

Setting	Type	Options / Values / Range
WDT Timeout Mode	Selection	Disabled [0x0] ■ default Minute [0x1] Second [0x2]
WDT Timeout Value	Numeric	Min 0x1 · Max 0xFF · Step 0x1 · Default 0x1E ■ default

S5 RTC Wake Settings (0x2739)

Setting	Type	Options / Values / Range
Wake system from S5	Selection	Disabled [0x0] ■ default Fixed Time [0x1] Dynamic Time [0x2]
Wake up hour	Numeric	Min 0x0 · Max 0x17 · Step 0x1 · Default 0x0 ■ default
Wake up minute	Numeric	Min 0x0 · Max 0x3B · Step 0x1 · Default 0x0 ■ default
Wake up second	Numeric	Min 0x0 · Max 0x3B · Step 0x1 · Default 0x0 ■ default
Wake up minute increase	Numeric	Min 0x1 · Max 0x5 · Step 0x1 · Default 0x1 ■ default

UEFI Variables Protection (0x273A)

Setting	Type	Options / Values / Range
Password protection of Runtime Variables	Selection	Enabled [0x1] ■ default Disabled [0x0]

CPU Configuration (0x273C)

Submenus Node 0 Information

Setting	Type	Options / Values / Range
PSS Support	Selection	Disabled [0x0] Enabled [0x1] ■ default ■ performance KEEP Enabled. Exposes ACPI P-states (_PSS) so the OS governor can scale frequency; required for normal turbo behavior.
PPC Adjustment	Selection	PState 0 [0x0] ■ default ■ performance TUNE PState 0. Pinning the OS to P0 (with Fixed SOC Pstate / APBDIS) holds max fabric/SoC clock for lowest latency and consistent boost; costs idle power. Leave Auto for efficiency.
PPC Adjustment	Selection	PState 0 [0x0] ■ default PState 1 [0x1] ■ performance TUNE PState 0. Pinning the OS to P0 (with Fixed SOC Pstate / APBDIS) holds max fabric/SoC clock for lowest latency and consistent boost; costs idle power. Leave Auto for efficiency.
PPC Adjustment	Selection	PState 0 [0x0] ■ default PState 1 [0x1] PState 2 [0x2] ■ performance TUNE PState 0. Pinning the OS to P0 (with Fixed SOC Pstate / APBDIS) holds max fabric/SoC clock for lowest latency and consistent boost; costs idle power. Leave Auto for efficiency.

Setting	Type	Options / Values / Range
PPC Adjustment	Selection	PState 0 [0x0] ■ default PState 1 [0x1] PState 2 [0x2] PState 3 [0x3] ■ performance TUNE PState 0. Pinning the OS to P0 (with Fixed SOC Pstate / APBDIS) holds max fabric/SoC clock for lowest latency and consistent boost; costs idle power. Leave Auto for efficiency.
PPC Adjustment	Selection	PState 0 [0x0] ■ default PState 1 [0x1] PState 2 [0x2] PState 3 [0x3] PState 4 [0x4] ■ performance TUNE PState 0. Pinning the OS to P0 (with Fixed SOC Pstate / APBDIS) holds max fabric/SoC clock for lowest latency and consistent boost; costs idle power. Leave Auto for efficiency.
PPC Adjustment	Selection	PState 0 [0x0] ■ default PState 1 [0x1] PState 2 [0x2] PState 3 [0x3] PState 4 [0x4] PState 5 [0x5] ■ performance TUNE PState 0. Pinning the OS to P0 (with Fixed SOC Pstate / APBDIS) holds max fabric/SoC clock for lowest latency and consistent boost; costs idle power. Leave Auto for efficiency.
PPC Adjustment	Selection	PState 0 [0x0] ■ default PState 1 [0x1] PState 2 [0x2] PState 3 [0x3] PState 4 [0x4] PState 5 [0x5] PState 6 [0x6] ■ performance TUNE PState 0. Pinning the OS to P0 (with Fixed SOC Pstate / APBDIS) holds max fabric/SoC clock for lowest latency and consistent boost; costs idle power. Leave Auto for efficiency.
PPC Adjustment	Selection	PState 0 [0x0] ■ default PState 1 [0x1] PState 2 [0x2] PState 3 [0x3] PState 4 [0x4] PState 5 [0x5] PState 6 [0x6] PState 7 [0x7] ■ performance TUNE PState 0. Pinning the OS to P0 (with Fixed SOC Pstate / APBDIS) holds max fabric/SoC clock for lowest latency and consistent boost; costs idle power. Leave Auto for efficiency.
NX Mode	Selection	Disabled [0x0] Enabled [0x1] ■ default ■ performance KEEP Enabled. No-Execute page protection; negligible perf cost and required by modern OSes.

AmiAgesaChipsetWrapper Configuration (0x2741)

Setting	Type	Options / Values / Range
PCR7 WA Support	Selection	Disabled [0x0] Enabled [0x1] Auto [0x2] ■ default
Enable Dual Vga Controllers	Selection	Disabled [0x0] ■ default Enabled [0x1]

AMI Graphic Output Protocol Policy (0x274B)

Setting	Type	Options / Values / Range
Output Select	Selection	Unknown Device [0x0] ■ default
Output Select	Selection	Unknown Device [0x0] ■ default Unknown Device [0x1]
Output Select	Selection	Unknown Device [0x0] ■ default Unknown Device [0x1] Unknown Device [0x2]
Output Select	Selection	Unknown Device [0x0] ■ default Unknown Device [0x1] Unknown Device [0x2] Unknown Device [0x3]
Output Select	Selection	Unknown Device [0x0] ■ default Unknown Device [0x1] Unknown Device [0x2] Unknown Device [0x3] Unknown Device [0x4]
Output Select	Selection	Unknown Device [0x0] ■ default Unknown Device [0x1] Unknown Device [0x2] Unknown Device [0x3] Unknown Device [0x4] Unknown Device [0x5]
Output Select	Selection	Unknown Device [0x0] ■ default Unknown Device [0x1] Unknown Device [0x2] Unknown Device [0x3] Unknown Device [0x4] Unknown Device [0x5] Unknown Device [0x6]
Output Select	Selection	Unknown Device [0x0] ■ default Unknown Device [0x1] Unknown Device [0x2] Unknown Device [0x3] Unknown Device [0x4] Unknown Device [0x5] Unknown Device [0x6] Unknown Device [0x7]

Debug Port Table (0x274D)

Setting	Type	Options / Values / Range
Debug Port Table	Selection	Disabled [0x0] ■ default Enabled [0x1]
Debug Port Table 2	Selection	Disabled [0x0] ■ default Enabled [0x1]

USB Configuration (0x2750)

Information fields USB Module Version, USB Controllers:, USB Devices:, USB hardware delays and time-outs:, Mass Storage Devices:

Setting	Type	Options / Values / Range
USB Support	Selection	Disabled [0x0] Enabled [0x1] ■ default
Legacy USB Support	Selection	Enabled [0x0] Disabled [0x1] ■ default Auto [0x2]
XHCI Legacy Support	Selection	Enabled [0x1] ■ default Disabled [0x0]
XHCI Hand-off	Selection	Enabled [0x1] ■ default Disabled [0x0]
EHCI Hand-off	Selection	Disabled [0x0] ■ default Enabled [0x1]
USB Mass Storage Driver Support	Selection	Disabled [0x0] Enabled [0x1] ■ default
USB transfer time-out	Selection	1 sec [0x1] 5 sec [0x5] 10 sec [0xA] 20 sec [0x14] ■ default
Device reset time-out	Selection	10 sec [0x0] 20 sec [0x1] ■ default 30 sec [0x2] 40 sec [0x3]
Device power-up delay	Selection	Auto [0x0] ■ default Manual [0x1]
Device power-up delay in seconds	Numeric	Min 0x1 · Max 0x28 · Step 0x1 · Default 0x5 ■ default
Mass Storage Device Emulation Type	Selection	Per-device emulation override, repeated once per detected USB mass-storage device. Options (identical for each): Auto [0x0] ■ default Floppy [0x1] Forced FDD [0x2] Hard Disk [0x3] CD-ROM [0x4]

Network Stack Configuration (0x2752)

Setting	Type	Options / Values / Range
Network Stack	Selection	Disabled [0x0] ■ default Enabled [0x1]

Setting	Type	Options / Values / Range
IPv4 PXE Support	Selection	Disabled [0x0] ■ default Enabled [0x1]
IPv4 HTTP Support	Selection	Disabled [0x0] ■ default Enabled [0x1]
IPv6 PXE Support	Selection	Disabled [0x0] ■ default Enabled [0x1]
IPv6 HTTP Support	Selection	Disabled [0x0] ■ default Enabled [0x1]
PXE boot wait time	Numeric	Min 0x0 · Max 0x5 · Step 0x1 · Default 0x0 ■ default
Media detect count	Numeric	Min 0x1 · Max 0x32 · Step 0x1 · Default 0x1 ■ default

Demo Board (0x2964)

Submenus PCI-E Port

Setting	Type	Options / Values / Range
Onboard PCIE LAN PXE ROM	Selection	Disabled [0x0] Enabled [0x1] ■ default

PCI-E Port (0x2966)

Setting	Type	Options / Values / Range
Pcie Port Control	Selection	Disabled [0x0] Enabled [0x1] ■ default ■ performance KEEP Enabled. Master enable for the per-port block below, not a link-speed or ASPM control. Disabling it hides the per-device options.
Device0	Selection	Disabled [0x0] Enabled [0x1] Auto [0xFF] ■ default
ASPM Mode Control(Device0)	Selection	Disabled [0x0] L0s Entry [0x1] L1 Entry [0x2] L0s And L1 Entry [0x3] Auto [0xFF] ■ default ■ performance TUNE PCIe link power management. Disabled = full link speed and lowest latency (best performance, higher idle power); Enabled (L1) = power saving with wake latency. Performance: Disabled.
Hotplug Mode Control(Device0)	Selection	Disabled [0x0] Hotplug Basic [0x1] Hotplug Server [0x2] Hotplug Enhanced [0x3] Hotplug Inboard [0x4] Auto [0xFF] ■ default
Device1	Selection	Disabled [0x0] Enabled [0x1] Auto [0xFF] ■ default

Setting	Type	Options / Values / Range
ASPM Mode Control(Device1)	Selection	Disabled [0x0] L0s Entry [0x1] L1 Entry [0x2] L0s And L1 Entry [0x3] Auto [0xFF] ■ default ■ performance TUNE PCIe link power management. Disabled = full link speed and lowest latency (best performance, higher idle power); Enabled (L1) = power saving with wake latency. Performance: Disabled.
Hotplug Mode Control(Device1)	Selection	Disabled [0x0] Hotplug Basic [0x1] Hotplug Server [0x2] Hotplug Enhanced [0x3] Hotplug Inboard [0x4] Auto [0xFF] ■ default
Device2	Selection	Disabled [0x0] Enabled [0x1] Auto [0xFF] ■ default
ASPM Mode Control(Device2)	Selection	Disabled [0x0] L0s Entry [0x1] L1 Entry [0x2] L0s And L1 Entry [0x3] Auto [0xFF] ■ default ■ performance TUNE PCIe link power management. Disabled = full link speed and lowest latency (best performance, higher idle power); Enabled (L1) = power saving with wake latency. Performance: Disabled.
Hotplug Mode Control(Device2)	Selection	Disabled [0x0] Hotplug Basic [0x1] Hotplug Server [0x2] Hotplug Enhanced [0x3] Hotplug Inboard [0x4] Auto [0xFF] ■ default
Device3	Selection	Disabled [0x0] Enabled [0x1] Auto [0xFF] ■ default
ASPM Mode Control(Device3)	Selection	Disabled [0x0] L0s Entry [0x1] L1 Entry [0x2] L0s And L1 Entry [0x3] Auto [0xFF] ■ default ■ performance TUNE PCIe link power management. Disabled = full link speed and lowest latency (best performance, higher idle power); Enabled (L1) = power saving with wake latency. Performance: Disabled.
Hotplug Mode Control(Device3)	Selection	Disabled [0x0] Hotplug Basic [0x1] Hotplug Server [0x2] Hotplug Enhanced [0x3] Hotplug Inboard [0x4] Auto [0xFF] ■ default
Device4	Selection	Disabled [0x0] Enabled [0x1] Auto [0xFF] ■ default

Setting	Type	Options / Values / Range
ASPM Mode Control(Device4)	Selection	Disabled [0x0] L0s Entry [0x1] L1 Entry [0x2] L0s And L1 Entry [0x3] Auto [0xFF] ■ default ■ performance TUNE PCIe link power management. Disabled = full link speed and lowest latency (best performance, higher idle power); Enabled (L1) = power saving with wake latency. Performance: Disabled.
Hotplug Mode Control(Device4)	Selection	Disabled [0x0] Hotplug Basic [0x1] Hotplug Server [0x2] Hotplug Enhanced [0x3] Hotplug Inboard [0x4] Auto [0xFF] ■ default
Device5	Selection	Disabled [0x0] Enabled [0x1] Auto [0xFF] ■ default
ASPM Mode Control(Device5)	Selection	Disabled [0x0] L0s Entry [0x1] L1 Entry [0x2] L0s And L1 Entry [0x3] Auto [0xFF] ■ default ■ performance TUNE PCIe link power management. Disabled = full link speed and lowest latency (best performance, higher idle power); Enabled (L1) = power saving with wake latency. Performance: Disabled.
Hotplug Mode Control(Device5)	Selection	Disabled [0x0] Hotplug Basic [0x1] Hotplug Server [0x2] Hotplug Enhanced [0x3] Hotplug Inboard [0x4] Auto [0xFF] ■ default
Device6	Selection	Disabled [0x0] Enabled [0x1] Auto [0xFF] ■ default
ASPM Mode Control(Device6)	Selection	Disabled [0x0] L0s Entry [0x1] L1 Entry [0x2] L0s And L1 Entry [0x3] Auto [0xFF] ■ default ■ performance TUNE PCIe link power management. Disabled = full link speed and lowest latency (best performance, higher idle power); Enabled (L1) = power saving with wake latency. Performance: Disabled.
Hotplug Mode Control(Device6)	Selection	Disabled [0x0] Hotplug Basic [0x1] Hotplug Server [0x2] Hotplug Enhanced [0x3] Hotplug Inboard [0x4] Auto [0xFF] ■ default
Device7	Selection	Disabled [0x0] Enabled [0x1] Auto [0xFF] ■ default

Setting	Type	Options / Values / Range
ASPM Mode Control(Device7)	Selection	Disabled [0x0] L0s Entry [0x1] L1 Entry [0x2] L0s And L1 Entry [0x3] Auto [0xFF] ■ default ■ performance TUNE PCIe link power management. Disabled = full link speed and lowest latency (best performance, higher idle power); Enabled (L1) = power saving with wake latency. Performance: Disabled.
Hotplug Mode Control(Device7)	Selection	Disabled [0x0] Hotplug Basic [0x1] Hotplug Server [0x2] Hotplug Enhanced [0x3] Hotplug Inboard [0x4] Auto [0xFF] ■ default

Chipset (0x2713)

Submenus South Bridge GFX Configuration North Bridge

South Bridge (0x2725)

Submenus SB USB Configuration SB MISC DEBUG Configuration 17113 GTRPRPI1001C · AMI Aptio V · AMD Strix Halo

SB USB Configuration (0x2726)

Submenus USB0 Ports USB1 Ports USB3 Ports USB4 Ports

USB0 Ports (0x2727)

Setting	Type	Options / Values / Range
XHCI0 Port 0	Selection	Disabled [0x0] Enabled [0x1] ■ default
XHCI0 Port 1	Selection	Disabled [0x0] Enabled [0x1] ■ default
XHCI0 2.0 Port 0	Selection	Disabled [0x0] Enabled [0x1] ■ default
XHCI0 2.0 Port 1	Selection	Disabled [0x0] Enabled [0x1] ■ default
XHCI0 2.0 Port 2	Selection	Disabled [0x0] Enabled [0x1] ■ default
XHCI0 2.0 Port 3	Selection	Disabled [0x0] Enabled [0x1] ■ default
XHCI0 2.0 Port 4	Selection	Disabled [0x0] Enabled [0x1] ■ default

USB1 Ports (0x2728)

Setting	Type	Options / Values / Range
XHCI1 Port 0	Selection	Disabled [0x0] Enabled [0x1] ■ default
XHCI1 2.0 Port 0	Selection	Disabled [0x0] Enabled [0x1] ■ default

USB3 Ports (0x2729)

Setting	Type	Options / Values / Range
XHCI3 Port 0	Selection	Disabled [0x0] Enabled [0x1] ■ default
XHCI3 2.0 Port 0	Selection	Disabled [0x0] Enabled [0x1] ■ default

USB4 Ports (0x272A)

Setting	Type	Options / Values / Range
XHCI4 Port 0	Selection	Disabled [0x0] Enabled [0x1] ■ default
XHCI4 2.0 Port 0	Selection	Disabled [0x0] Enabled [0x1] ■ default

SB MISC DEBUG Configuration (0x272B)

Setting	Type	Options / Values / Range
HPET ACPI table In SB	Selection	Disabled [0x0] Enabled [0x1] Auto [0xFF] ■ default ■ performance TUNE HPET: leave Enabled (Auto). Modern AMD platforms use invariant TSC; forcing HPET off rarely helps and can break timing-sensitive software.

North Bridge (0x273F)

Submenus Socket 0 Information

Security (0x2714)

Submenus Secure Boot ■ Secure Flash Update

Information fields Minimum length, Maximum length

Setting	Type	Options / Values / Range
Administrator Password	Password	Password entry
User Password	Password	Password entry

Secure Boot (0x2815)

Submenus Restore Factory Keys Reset To Setup Mode Expert Key Management

Information fields System Mode ■■

Setting	Type	Options / Values / Range
Secure Boot	Selection	Disabled [0x0] ■ default Enabled [0x1]
Secure Boot Mode	Selection	Standard [0x0] Custom [0x1] ■ default

Expert Key Management (0x281F)

Submenus Restore Factory Keys ■ Reset To Setup Mode ■ Enroll Efi Image ■ Export Secure Boot variables

Information fields Vendor Keys

Setting	Type	Options / Values / Range
Factory Key Provision	Selection	Disabled [0x0] ■ default Enabled [0x1]

Secure Flash Update (0x282E)

Information fields Signed FW Update, FW Key File, FW Key Type, FW Update Methods, FW Revision Rollback, Flash Write Protect

Boot (0x2715)

Submenus UEFI NVME Drive BBS Priorities UEFI Hard Disk Drive BBS Priorities UEFI CDROM/DVD Drive BBS Priorities ■■■ UEFI SD Drive BBS Priorities UEFI USB Drive BBS Priorities UEFI NETWORK Drive BBS Priorities UEFI Application Boot Priorities

Setting	Type	Options / Values / Range
Setup Prompt Timeout	Numeric	Min 0x1 · Max 0xFFFF · Step 0x1 · Default 0x1 ■ default
Bootup NumLock State	Selection	On [0x1] ■ default Off [0x0]
Fast Boot	Selection	Disabled [0x0] ■ default Enabled [0x1] ■ performance TUNE Enabled for faster POST (skips some device init). Disable temporarily if you need full enumeration or to enter Setup reliably.
SATA Support	Selection	Last Boot SATA Devices Only [0x0] ■ default All SATA Devices [0x1] ■ performance KEEP Boot-time SATA enumeration scope, not a runtime speed control. Last Boot SATA Devices Only re-scans only known ports (faster POST); All SATA Devices for hot-swap.
NVMe Support	Selection	Disabled [0x0] Enabled [0x1] ■ default ■ performance KEEP Keep enabled for the storage/devices you use. Enabling NVMe support is required to boot/perform from NVMe; no downside.
UFS Support	Selection	Disabled [0x0] Enabled [0x1] ■ default ■ performance KEEP Leave enabled if you use UFS storage; no runtime performance impact. Disable only to hide an unused UFS controller. No downside to leaving on.
VGA Support	Selection	Auto [0x0] ■ default EFI Driver [0x1] ■ performance KEEP Video BIOS / GOP init policy, not performance. Auto loads legacy VGA when present; EFI Driver forces UEFI GOP. Leave Auto unless a GPU needs pure-UEFI output.
USB Support	Selection	Disabled [0x0] Full Initial [0x1] ■ default Partial Initial [0x2]
PS2 Devices Support	Selection	Disabled [0x0] Enabled [0x1] ■ default ■ performance KEEP Legacy PS/2 keyboard/mouse init, not performance. Harmless to leave enabled; this mini-PC has no PS/2 port, so it has no runtime effect. Leave default.

Setting	Type	Options / Values / Range
Network Stack Driver Support	Selection	Disabled [0x0] ■ default Enabled [0x1]
Redirection Support	Selection	Disabled [0x0] ■ default Enabled [0x1]
Driver Option #%d	Selection	(populated at runtime) [0x0] ■ default (populated at runtime) [0x1]
Boot mode select	Selection	LEGACY [0x0] UEFI [0x1] ■ default ■ performance KEEP UEFI (not Legacy/CSM). UEFI is required for Resizable BAR and modern features; CSM blocks Above-4G/ReBAR.
Boot Option #1	Selection	NVME [0x0] ■ default Hard Disk [0x1] CD/DVD [0x2] SD [0x3] USB Device [0x4] Network [0x5] UEFI AP [0x6] Disabled [0x7]
Boot Option #2	Selection	NVME [0x0] Hard Disk [0x1] ■ default CD/DVD [0x2] SD [0x3] USB Device [0x4] Network [0x5] UEFI AP [0x6] Disabled [0x7]
Boot Option #3	Selection	NVME [0x0] Hard Disk [0x1] CD/DVD [0x2] ■ default SD [0x3] USB Device [0x4] Network [0x5] UEFI AP [0x6] Disabled [0x7]
Boot Option #4	Selection	NVME [0x0] Hard Disk [0x1] CD/DVD [0x2] SD [0x3] ■ default USB Device [0x4] Network [0x5] UEFI AP [0x6] Disabled [0x7]
Boot Option #5	Selection	NVME [0x0] Hard Disk [0x1] CD/DVD [0x2] SD [0x3] USB Device [0x4] ■ default Network [0x5] UEFI AP [0x6] Disabled [0x7]

Setting	Type	Options / Values / Range
Boot Option #6	Selection	NVME [0x0] Hard Disk [0x1] CD/DVD [0x2] SD [0x3] USB Device [0x4] Network [0x5] ■ default UEFI AP [0x6] Disabled [0x7]
Boot Option #7	Selection	NVME [0x0] Hard Disk [0x1] CD/DVD [0x2] SD [0x3] USB Device [0x4] Network [0x5] UEFI AP [0x6] ■ default Disabled [0x7]
Boot Option #1	Selection	Hard Disk [0x0] ■ default NVME [0x1] CD/DVD [0x2] SD [0x3] USB Device [0x4] Network [0x5] Disabled [0x6]
Boot Option #2	Selection	Hard Disk [0x0] NVME [0x1] ■ default CD/DVD [0x2] SD [0x3] USB Device [0x4] Network [0x5] Disabled [0x6]
Boot Option #3	Selection	Hard Disk [0x0] NVME [0x1] CD/DVD [0x2] ■ default SD [0x3] USB Device [0x4] Network [0x5] Disabled [0x6]
Boot Option #4	Selection	Hard Disk [0x0] NVME [0x1] CD/DVD [0x2] SD [0x3] ■ default USB Device [0x4] Network [0x5] Disabled [0x6]
Boot Option #5	Selection	Hard Disk [0x0] NVME [0x1] CD/DVD [0x2] SD [0x3] USB Device [0x4] ■ default Network [0x5] Disabled [0x6]

Setting	Type	Options / Values / Range
Boot Option #6	Selection	Hard Disk [0x0] NVME [0x1] CD/DVD [0x2] SD [0x3] USB Device [0x4] Network [0x5] ■ default Disabled [0x6]

AMD CBS (Common BIOS Settings)

51 forms · 318 settings

AMD CBS (0x7000)

Submenus CPU Common Options ■ DF Common Options ■ UMC Common Options ■ NBIO Common Options ■ FCH

Information fields AMD CBS Revision Number Common Options ■ SMU Common Options ■ SOC Miscellaneous Control

Setting	Type	Options / Values / Range
Combo CBS	Numeric	Min 0x0 · Max 0xFF · Default 0xFE ■ default

CPU Common Options (0x7001)

Submenus Performance ■ Prefetcher settings ■ Core Watchdog

Setting	Type	Options / Values / Range
REP-MOV/STOS Streaming	Selection	Disabled [0x0] Enabled [0x1] ■ default ■ performance KEEP Enabled (Auto). Streaming string ops improve large memcpy/memset bandwidth.
RedirectForReturnDis	Selection	Auto [0xFF] ■ default 1 [0x1] 0 [0x0]
Platform First Error Handling	Selection	Enabled [0x1] Disabled [0x0] Auto [0x3] ■ default
Core Performance Boost	Selection	Disabled [0x0] Auto [0x1] ■ default ■ performance KEEP Auto (boost enabled). Allows cores to boost above base clock; the single biggest stock throughput lever. Leave it alone.
Global C-state Control	Selection	Disabled [0x0] ■ default Enabled [0x1] Auto [0x3] ■ performance TUNE Enabled for daily/efficiency. Disable only to chase last-percent latency consistency (raises idle power/heat, reduces boost headroom).
Opcache grayout flag	Selection	0 [0x0] ■ default 1 [0x1] Display [0x2]
Opcache Control	Selection	Disabled [0x1] Enabled [0x0] Auto [0xFF] ■ default ■ performance KEEP Enabled (Auto). The decoded-op cache improves front-end throughput and efficiency; leave on.
SEV ASID Count	Selection	253 ASIDs [0x0] 509 ASIDs [0x1] Auto [0x3] ■ default
SEV-ES ASID Space Limit Control	Selection	Auto [0x0] ■ default Manual [0x1]
SEV-ES ASID Space Limit	Numeric	Min 0x1 · Max 0x1FE · Default 0x1 ■ default

Setting	Type	Options / Values / Range
Streaming Stores Control	Selection	Disabled [0x1] Enabled [0x0] Auto [0xFF] ■ default ■ performance KEEP Enabled (Auto). Non-temporal stores help bandwidth-bound workloads.
Local APIC Mode	Selection	Compatibility [0x0] xAPIC [0x1] x2APIC [0x2] Auto [0xFF] ■ default ■ performance KEEP Auto/x2APIC. x2APIC improves interrupt scaling on high-core-count parts; Auto selects correctly.
ACPI _CST C1 Declaration	Selection	Disabled [0x0] Enabled [0x1] Auto [0x3] ■ default ■ performance KEEP Enabled (Auto). Lets the OS use C1 for efficiency without measurable throughput loss.
MCA error thresh enable	Selection	False [0x0] True [0x1] Auto [0xFF] ■ default
MCA error thresh count	Numeric	Min 0x1 · Max 0xFFFF · Default 0xFF5 ■ default
SMU and PSP Debug Mode	Selection	Disabled [0x0] Enabled [0x1] Auto [0x3] ■ default
Enhanced REP MOVSB/STOSB (ERSM)	Selection	Disabled [0x0] Enabled [0x1] Auto [0xFF] ■ default ■ performance KEEP Enabled (Auto). Faster string-copy microcode path.
Log Transparent Errors	Selection	Auto [0x3] ■ default Disabled [0x0] Enabled [0x1]
AVX512	Selection	Disabled [0x0] Enabled [0x1] Auto [0xFF] ■ default ■ performance KEEP Enabled (Auto). AVX-512 accelerates vectorized/AI workloads on Zen 5; only disable for thermal-limited edge cases.
MONITOR and MWAIT disable	Selection	Enabled [0x1] Disabled [0x0] Auto [0xFF] ■ default ■ performance KEEP Keep MWAIT enabled (option Disabled = false). MWAIT-based idle is more efficient and lower-latency than polling.
SVM Lock	Selection	Enabled [0x1] Disabled [0x0] Auto [0xFF] ■ default
SVM Enable	Selection	Enabled [0x1] Disabled [0x0] Auto [0xFF] ■ default ■ performance TUNE Enabled if you run VMs/containers/WSL2 (AMD-V); a few-percent overhead otherwise. Most users: Enabled.

Setting	Type	Options / Values / Range
Corrector Branch Predictor	Selection	Disable [0x0] ■ default Enable [0x1] ■ performance KEEP Enabled (Auto). Improves branch prediction accuracy; leave on.
PAUSE Delay	Selection	Auto [0xFF] ■ default Disable [0x0] 16 cycles [0x1] 32 cycles [0x2] 64 cycles [0x3] 128 cycles [0x4]

Performance (0x7008)

Submenus Custom Core Pstates

Setting	Type	Options / Values / Range
OC Mode	Selection	Normal Operation [0x0] OC1 [0x1] OC2 [0x2] OC3 [0x3] Customized [0x5] ■ default ■ performance TUNE Normal Operation unless you have a deliberate tuning plan. 'Customized' unlocks manual P-states and is expert-only.

Custom Core Pstates (0x700F)

Submenus Decline Accept

Information fields WARNING — DAMAGE CAUSED BY USE OF YOUR AMD PROCESSOR OUTSIDE OF SPECIFICATION OR IN EXCESS OF FACTORY SETTINGS ARE NOT COVERED UNDER YOUR AMD PRODUCT WARRANTY AND MAY NOT BE COVERED BY YOUR SYSTEM MANUFACTURER'S WARRANTY., Operating your AMD processor outside of specification or in excess of factory settings ■

Accept (0x7011)

Submenus LPDDR SPD Timing LPDDR Non-SPD Timing

Setting	Type	Options / Values / Range
Custom Pstate0	Selection	Custom [0x1] Auto [0x2] ■ default ■ performance TUNE Expert-only manual P0 definition; leave default unless you know your stable freq/VID pair.
Pstate0 Freq (MHz)	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ default ■ performance TUNE Manual P0 frequency — only set with a validated freq/VID; PBO+CO is safer for gains.
Pstate0 VID	Numeric	Min 0x0 · Max 0x1FF · Default 0x0 ■ default ■ performance TUNE Manual P0 voltage ID — pair carefully with the frequency; wrong values prevent boot.

Prefetcher settings (0x7009)

Setting	Type	Options / Values / Range
L1 Stream HW Prefetcher	Selection	Disable [0x0] Enable [0x1] Auto [0x3] ■ default ■ performance KEEP Enabled (Auto). Hardware prefetch hides memory latency; broadly positive. Disable only if profiling a specific regression.
L2 Stream HW Prefetcher	Selection	Disable [0x0] Enable [0x1] Auto [0x3] ■ default ■ performance KEEP Enabled (Auto). Improves memory-bound throughput; leave on.
L2 Up/Down Prefetcher	Selection	Disable [0x0] Enable [0x1] Auto [0x3] ■ default ■ performance KEEP Enabled (Auto). Bidirectional stride prefetch; generally beneficial.

Core Watchdog (0x700A)

Setting	Type	Options / Values / Range
Core Watchdog Timer Enable	Selection	Disabled [0x0] Enabled [0x1] Auto [0x3] ■ default
Core Watchdog Timer Interval	Selection	21.461s [0x900] 10.730s [0x800] 5.364s [0x0] 2.681s [0x100] 1.340s [0x200] 669.41ms [0x300] 334.05ms [0x400] 166.37ms [0x500] 82.53ms [0x600] 40.61ms [0x700] 20.970ms [0x901] 10.484ms [0x801] 5.241ms [0x1] 2.620ms [0x101] 1.309ms [0x201] 654.08us [0x301] 326.4us [0x401] 162.56us [0x501] 80.64us [0x601] 39.68us [0x701] Auto [0xFFFF] ■ default
Core Watchdog Timer Severity	Selection	No Error [0x0] Transparent [0x1] Corrected [0x2] Deferred [0x3] Uncorrected [0x4] Fatal [0x5] Auto [0xFF] ■ default

DF Common Options (0x7002)

Submenus Memory Addressing

Setting	Type	Options / Values / Range
DF Watchdog Timer Interval	Selection	Auto [0xFF] ■ default 41 ms [0x0] 166 ms [0x1] 334 ms [0x2] 669 ms [0x3] 1.34 seconds [0x4] 2.68 seconds [0x5] 5.36 seconds [0x6]
Disable DF to external downstream IP SyncFloodPropagation	Selection	Sync flood disabled [0x1] Sync flood enabled [0x0] Auto [0xFF] ■ default
Sync Flood Propagation to DF Components	Selection	Sync flood disabled [0x1] Sync flood enabled [0x0] Auto [0xFF] ■ default
Freeze DF module queues on error	Selection	Disabled [0x0] Enabled [0x1] Auto [0x3] ■ default
CC6 memory region encryption	Selection	Disabled [0x0] Enabled [0x1] Auto [0x3] ■ default
System probe filter	Selection	Disabled [0x0] Enabled [0x1] Auto [0x3] ■ default ■ performance KEEP Auto/Enabled. Probe filter reduces coherency traffic — helps multi-core memory performance.
Disable DF sync flood propagation	Selection	Sync flood disabled [0x1] Sync flood enabled [0x0] Auto [0x3] ■ default
DF Cstates	Selection	Disabled [0x0] Enabled [0x1] Auto [0xFF] ■ default ■ performance TUNE Disabled for lowest-latency / sustained all-core work (keeps Infinity Fabric out of low-power state); Enabled (Auto) for best idle efficiency.

Memory Addressing (0x7015)

Setting	Type	Options / Values / Range
Memory interleaving	Selection	Disabled [0x0] Auto [0x7] ■ default ■ performance KEEP Auto. Channel interleaving maximises memory bandwidth; Disabled costs throughput on every workload.
Memory interleaving size	Selection	256 Bytes [0x0] 512 Bytes [0x1] 1 KB [0x2] 2 KB [0x3] Auto [0x7] ■ default ■ performance KEEP Auto. Firmware picks the interleave granularity that suits the soldered config.

Setting	Type	Options / Values / Range
DRAM map inversion	Selection	Disabled [0x0] Enabled [0x1] Auto [0x3] ■ default
Location of private memory regions	Selection	Distributed [0x0] Consolidated [0x1] Consolidated to 1st DRAM pair [0x2] Auto [0xFF] ■ default
NUMA Nodes	Selection	NPS1 [0x1] ■ default NPS2 [0x2] NPS4 [0x4] ■ performance KEEP NPS1 (single node). Strix Halo presents one unified memory domain across both CCDs on a 256-bit LPDDR5X interface; NPS2/NPS4 gain nothing here and only complicate scheduling.

UMC Common Options (0x7003)

Submenus LPDDR Options

LPDDR Options (0x7019)

Submenus LPDDR Timing Configuration LPDDR Controller Configuration LPDDR Bus Configuration LPDDR RAS ■■■ LPDDR Security
LPDDR Addressing Options LPDDR Training Options LPDDR Memory MBIST

LPDDR Timing Configuration (0x701A)

Submenus Decline Accept

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Accept (0x7023)

Submenus LPDDR SPD Timing LPDDR Non-SPD Timing

Setting	Type	Options / Values / Range
Active Memory Timing Settings	Selection	Auto [0xFF] ■ default Enabled [0x1]
Maximum Memory Data Clock Speed	Selection	Auto [0xFFFF] ■ default 1600MT/s [0x640] 3200MT/s [0xC80] 4267MT/s [0x10AB] 5500MT/s [0x157C] 6400MT/s [0x1900] 7500MT/S [0x1D4C] 8000MT/S [0x1F40] ■ performance TUNE Leave at the SoC default (LPDDR5X-8000 is soldered and pre-tuned). Forcing higher than the validated rate risks training failure with no headroom on soldered memory.

LPDDR SPD Timing (0x7024)

Setting	Type	Options / Values / Range
Trcpage Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Trcpage	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Tcwl Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Tcwl	Numeric	Min 0x9 · Max 0x16 · Default 0x9 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Tcl Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Tcl	Numeric	Min 0x8 · Max 0x21 · Default 0x8 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Trcdrd Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Trcdrd	Numeric	Min 0x8 · Max 0x1B · Default 0x8 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Trcdwr Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.

Setting	Type	Options / Values / Range
Trcdwr	Numeric	Min 0x8 · Max 0x1B · Default 0x8 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Trp Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Trp	Numeric	Min 0x8 · Max 0x1B · Default 0x8 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TrpPb Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TrpPb	Numeric	Min 0x8 · Max 0x1B · Default 0x8 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Trfc Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Trfc	Numeric	Min 0x3C · Max 0x3DE · Default 0x138 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TrfcPb Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TrfcPb	Numeric	Min 0x3C · Max 0x3DE · Default 0x138 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.

LPDDR Non-SPD Timing (0x7025)

Setting	Type	Options / Values / Range
Tras Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Tras	Numeric	Min 0x15 · Max 0x3A · Default 0x15 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Trc Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Trc	Numeric	Min 0x1D · Max 0x87 · Default 0x39 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TrcPb Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TrcPb	Numeric	Min 0x1D · Max 0x87 · Default 0x39 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TrrdS Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TrrdS	Numeric	Min 0x4 · Max 0xC · Default 0x4 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TrrdL Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.

Setting	Type	Options / Values / Range
TrrdL	Numeric	Min 0x4 · Max 0xC · Default 0x4 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Tfaw Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Tfaw	Numeric	Min 0x6 · Max 0x36 · Default 0x1A ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TwtrS Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TwtrS	Numeric	Min 0x2 · Max 0xE · Default 0x2 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TwtrL Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TwtrL	Numeric	Min 0x2 · Max 0xE · Default 0x2 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Twr Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Twr	Numeric	Min 0xA · Max 0x51 · Default 0x12 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.

Setting	Type	Options / Values / Range
Trtp Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Trtp	Numeric	Min 0x5 · Max 0xE · Default 0x5 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Tcke Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Tcke	Numeric	Min 0x1 · Max 0x1F · Default 0x1 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TrdrdScL Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TrdrdScL	Numeric	Min 0x1 · Max 0xF · Default 0x3 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TrdrdSc Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TrdrdSc	Numeric	Min 0x1 · Max 0xF · Default 0x1 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TrdrdSd Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.

Setting	Type	Options / Values / Range
TrdrdSd	Numeric	Min 0x1 · Max 0xF · Default 0x1 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TrdrdDd Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TrdrdDd	Numeric	Min 0x1 · Max 0xF · Default 0x1 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TwrrwrScL Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TwrrwrScL	Numeric	Min 0x1 · Max 0x3F · Default 0x3 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TwrrwrSc Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TwrrwrSc	Numeric	Min 0x1 · Max 0xF · Default 0x1 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TwrrwrSd Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TwrrwrSd	Numeric	Min 0x1 · Max 0xF · Default 0x1 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.

Setting	Type	Options / Values / Range
TwrrrDd Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TwrrrDd	Numeric	Min 0x1 · Max 0xF · Default 0x1 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Twrrd Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Twrrd	Numeric	Min 0x1 · Max 0xF · Default 0x1 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Trdwr Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Trdwr	Numeric	Min 0x1 · Max 0x1F · Default 0x1 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.

LPDDR Controller Configuration (0x701B)

Submenus LPDDR Power Options

Setting	Type	Options / Values / Range
LPDDR Refresh Mode	Selection	Auto [0xFF] ■ default All Banks [0x1] Per Bank [0x2] ■ performance KEEP Auto. Refresh policy is tuned for the soldered LPDDR5X; leave default.
LPDDR RFM	Selection	Disabled [0x0] Enabled [0x1] Auto [0xFF] ■ default ■ performance KEEP Auto. Refresh-management feature — leave to firmware.
WCK Always On	Selection	Disabled [0x0] Enabled [0x1] Auto [0xFF] ■ default

Setting	Type	Options / Values / Range
DVFSC Mode	Selection	Disabled [0x0] Enabled [0x1] Auto [0xFF] ■ default ■ performance KEEP Auto. Dynamic memory voltage/freq scaling — leave to firmware.
RRW Memory Test Control	Selection	Auto [0xFF] ■ default Enabled [0x1] Disabled [0x0]

LPDDR Power Options (0x7026)

Setting	Type	Options / Values / Range
Power Down Enable	Selection	Disabled [0x0] Enabled [0x1] Auto [0xFF] ■ default ■ performance TUNE Disabled can trim memory latency slightly under load; Enabled (Auto) saves idle power. Default is fine for most.
Phy Low Power Disable	Selection	Auto [0xFF] ■ default 0 [0x0] 1 [0x1]

LPDDR Bus Configuration (0x701C)

Setting	Type	Options / Values / Range
CA drive strengths	Selection	Auto [0xFF] ■ default 120.0 Ohm [0x78] 60.0 Ohm [0x3C] 40.0 Ohm [0x28]
CK drive strengths	Selection	Auto [0xFF] ■ default 120.0 Ohm [0x78] 60.0 Ohm [0x3C] 40.0 Ohm [0x28]
DQ drive strengths	Selection	Auto [0xFF] ■ default 120.0 Ohm [0x78] 60.0 Ohm [0x3C] 40.0 Ohm [0x28]
DQS drive strengths	Selection	Auto [0xFF] ■ default 120.0 Ohm [0x78] 60.0 Ohm [0x3C] 40.0 Ohm [0x28]
WCK drive strengths	Selection	Auto [0xFF] ■ default 120.0 Ohm [0x78] 60.0 Ohm [0x3C] 40.0 Ohm [0x28]
Processor CA ODT impedance	Selection	Auto [0xFF] ■ default 120.0 Ohm [0x78] 60.0 Ohm [0x3C] 40.0 Ohm [0x28] Disable [0x0]

Setting	Type	Options / Values / Range
Processor CK ODT impedance	Selection	Auto [0xFF] ■ default 120.0 Ohm [0x78] 60.0 Ohm [0x3C] 40.0 Ohm [0x28] Disable [0x0]
Processor CS ODT impedance	Selection	Auto [0xFF] ■ default 120.0 Ohm [0x78] 60.0 Ohm [0x3C] 40.0 Ohm [0x28] Disable [0x0]
Processor DQ ODT impedance	Selection	Auto [0xFF] ■ default 120.0 Ohm [0x78] 60.0 Ohm [0x3C] 40.0 Ohm [0x28] Disable [0x0]
Processor DQS ODT impedance	Selection	Auto [0xFF] ■ default 120.0 Ohm [0x78] 60.0 Ohm [0x3C] 40.0 Ohm [0x28] Disable [0x0]
Processor WCK ODT impedance	Selection	Auto [0xFF] ■ default 120.0 Ohm [0x78] 60.0 Ohm [0x3C] 40.0 Ohm [0x28] Disable [0x0]
Dram CA ODT impedance	Selection	Auto [0xFF] ■ default Disable [0x0] RZQ/1 [0x1] RZQ/2 [0x2] RZQ/3 [0x3] RZQ/4 [0x4] RZQ/5 [0x5] RZQ/6 [0x6]
Dram CS ODT impedance	Selection	Auto [0xFF] ■ default Disable [0x0] RZQ/1 [0x1] RZQ/2 [0x2] RZQ/3 [0x3]
Dram DQ ODT impedance	Selection	Auto [0xFF] ■ default Disable [0x0] RZQ/1 [0x1] RZQ/2 [0x2] RZQ/3 [0x3] RZQ/4 [0x4] RZQ/5 [0x5] RZQ/6 [0x6]

Setting	Type	Options / Values / Range
Dram WCK ODT impedance	Selection	Auto [0xFF] ■ default Disable [0x0] RZQ/1 [0x1] RZQ/2 [0x2] RZQ/3 [0x3] RZQ/4 [0x4] RZQ/5 [0x5] RZQ/6 [0x6]
Dram Non-Target ODT impedance	Selection	Auto [0xFF] ■ default Disable [0x0] RZQ/1 [0x1] RZQ/2 [0x2] RZQ/3 [0x3] RZQ/4 [0x4] RZQ/5 [0x5] RZQ/6 [0x6]
Dram Pull-Down drive strengths	Selection	Auto [0xFF] ■ default Disable [0x0] RZQ/1 [0x1] RZQ/2 [0x2] RZQ/3 [0x3] RZQ/4 [0x4] RZQ/5 [0x5] RZQ/6 [0x6]

LPDDR RAS (0x701D)

Setting	Type	Options / Values / Range
DRAM Read Link ECC Enable	Selection	Disabled [0x0] Enabled [0x1] Auto [0xFF] ■ default ■ performance TUNE Link ECC adds reliability with a small bandwidth cost. Disabled for maximum memory throughput; Enabled for data integrity. Default is fine.
DRAM Write Link ECC Enable	Selection	Disabled [0x0] Enabled [0x1] Auto [0xFF] ■ default ■ performance TUNE Link ECC adds reliability with a small bandwidth cost. Disabled for maximum memory throughput; Enabled for data integrity. Default is fine.
Disable Memory Error Injection	Selection	False [0x0] True [0x1] Auto [0xFF] ■ default

LPDDR Security (0x701E)

Setting	Type	Options / Values / Range
TSME	Selection	Auto [0xFF] ■ default Enabled [0x1] Disabled [0x0] ■ performance TUNE Transparent SME (full-memory encryption) adds a small bandwidth/latency cost. Disabled for maximum memory performance; Enabled for at-rest RAM encryption.

Setting	Type	Options / Values / Range
Data Scramble	Selection	Enabled [0x1] Disabled [0x0] Auto [0xFF] ■ default

LPDDR Addressing Options (0x701F)

Setting	Type	Options / Values / Range
Chip Select Interleaving	Selection	Disabled [0x0] Auto [0xFF] ■ default ■ performance KEEP Auto. Interleaving across chip selects aids bandwidth; leave it on.
Bank Swap	Selection	Disabled [0x0] Enabled [0x1] Auto [0xFF] ■ default ■ performance KEEP Auto. Bank remapping for access spreading — leave to firmware.
BankGroup Swap	Selection	Disabled [0x0] Enabled [0x2] Auto [0xFF] ■ default ■ performance KEEP Auto. Bank-group swap can improve access parallelism; leave default.
Address Hash Bank	Selection	Disabled [0x0] Enabled [0x1] Auto [0xFF] ■ default ■ performance KEEP Auto/Enabled. Address hashing reduces bank conflicts — beneficial for bandwidth.
Address Hash CS	Selection	Auto [0xFF] ■ default Enabled [0x1] Disabled [0x0] ■ performance KEEP Auto/Enabled. Chip-select hashing spreads accesses — leave on.

LPDDR Training Options (0x7020)

Setting	Type	Options / Values / Range
DFE Read Training	Selection	Auto [0xFF] ■ default Enable [0x1] Disable [0x0]
DFE Write Training	Selection	Auto [0xFF] ■ default Enable [0x1] Disable [0x0]

LPDDR Memory MBIST (0x7021)

Submenu LPDDR Data Eye

Setting	Type	Options / Values / Range
MBIST Enable	Selection	Disabled [0x0] Enabled [0x1] Auto [0xFF] ■ default

Setting	Type	Options / Values / Range
MBIST Test Mode	Selection	Interface Mode [0x0] Data Eye Mode [0x1] Both [0x2] Auto [0xFF] ■ default
MBIST Aggressors	Selection	Disabled [0x0] Enabled [0x1] Auto [0xFF] ■ default

LPDDR Data Eye (0x7029)

Setting	Type	Options / Values / Range
Pattern Select	Selection	Auto [0xFF] ■ default PRBS [0x0] SSO [0x1] Both [0x2]
Pattern Length Select	Selection	Auto [0xFF] ■ default Manual [0x1]
Pattern Length	Numeric	Min 0x3 · Max 0xC · Default 0x3 ■ default
Aggressor Channel	Selection	Auto [0xFF] ■ default Disabled [0x0] 1 Aggressor Channel [0x1] 3 Aggressor Channels [0x3] 7 Aggressor Channels [0x7]
Read Voltage Sweep Step Size	Selection	Auto [0xFF] ■ default 1 [0x0] 2 [0x1] 4 [0x4]
Read Timing Sweep Step Size	Selection	Auto [0xFF] ■ default 1 [0x1] 2 [0x2] 4 [0x4]
Write Voltage Sweep Step Size	Selection	Auto [0xFF] ■ default 1 [0x0] 2 [0x1] 4 [0x4]
Write Timing Sweep Step Size	Selection	Auto [0xFF] ■ default 1 [0x1] 2 [0x2] 4 [0x4]

NBIO Common Options (0x7004)

Submenus [GFX Configuration](#) ■ [Audio Configuration](#)

Setting	Type	Options / Values / Range
IOMMU	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default ■ performance TUNE Disabled yields roughly 6% more iGPU memory-read bandwidth and slightly lower latency. Keep Enabled if you use VFIO/GPU passthrough, the NPU, or reliable suspend - all three depend on it.

Setting	Type	Options / Values / Range
DMAr Support	Selection	Auto [0xF] ■ default Enabled [0x1] Disabled [0x0]
DMA Protection	Selection	Auto [0xF] ■ default Enabled [0x1] Disabled [0x0]
SCPC attribute control	Selection	0 [0x0] 1 [0x1] 2 [0x2] 3 [0x3] Customized [0xFF] ■ default
PCIe ARI Support	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default ■ performance KEEP Enabled (Auto). Alternative Routing-ID improves multi-function PCIe device support; leave on.
PCIe loopback Mode	Selection	Auto [0xF] ■ default Disabled [0x0] Enabled [0x1]
PSPP Policy	Selection	Auto [0xF] ■ default Disabled [0x0] Balanced [0x2] ■ performance TUNE Balanced (Auto) for daily; set to Disabled/Performance to keep PCIe links at full speed (lower latency, higher idle power).

GFX Configuration (0x702C)

Setting	Type	Options / Values / Range
iGPU Configuration	Selection	Auto [0xF] iGPU Disabled [0x0] UMA_SPECIFIED [0x1] ■ default UMA_AUTO [0x2] UMA_GAME_OPTIMIZED [0x3] UMA_WORKSTATION_OPTIMIZED [0x4] ■ performance KEEP UMA_SPECIFIED (default). The Radeon 8060S is the only graphics on this APU - iGPU Disabled leaves the machine headless.
iGPU Mem Bar Configuration	Selection	Auto [0xF] ■ default LegacyGfxBAR [0x1] LargeBAR [0x2] ResizableBAR [0x3] ■ performance KEEP Leave Auto. Large iGPU BAR pairs with Above-4G/ReBAR for full-aperture access.
UMA Version	Selection	Legacy [0x0] Non-Legacy [0x1] Auto [0xF] ■ default

Setting	Type	Options / Values / Range
UMA Frame buffer Size	Selection	512M [0x200] 4G [0x1000] 8G [0x2000] 16G [0x4000] 32G [0x8000] 64G [0x10000] 96G [0x18000] ■ default ■ performance TUNE Workload-dependent. Windows gaming/AI: large carve-out (e.g. 64-96 GB) for guaranteed VRAM. Linux LLM (Vulkan/GTT) or general use: minimum (512 MB) so the OS sees full RAM — GTT still gives the GPU the whole pool with no speed loss.
GPU Host Translation Cache	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default ■ performance KEEP Enabled (Auto). Caches GPU address translations — helps iGPU memory-access latency.
TCON INSTANT ON LOGO	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default
Dedicated Graphics Memory Auto Level	Selection	UmaSizeMinimum [0x0] ■ default UmaSizeMedium [0x1] UmaSizeHigh [0x2]
Remaining System Memory	Selection	RemainGB [0x0] ■ default

Audio Configuration (0x702D)

Setting	Type	Options / Values / Range
NB Azalia	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default
Audio IOs	Selection	Auto [0xFF] ■ default HDA(3SDI) [0x1] HDA(1SDI) + SW0(1MDATA) [0x2] SW0(4MDATA) + SW1(1MDATA) [0x3] HDA(3SDI) + PDM(2CH) [0x4] HDA(1SDI) + PDM(6CH) [0x5] HDA(1SDI) + SW0(1MDATA) + PDM(2CH) [0x6] SW0(4MDATA) + PDM(6CH) [0x7] SW0(4MDATA) + SW1(1MDATA) + PDM(2CH) [0x8] 3I2S + 1 REFCLK + 1 INTR [0x9] HDA(3SDI) + PDM(6CH) + I2S [0xA] HDA(3SDI) + PDM(8CH) [0xB] HDA(1SDI) + SW0(1MDATA) + PDM(6CH) + I2S [0xC] SW0(4MDATA) + SW1(1MDATA) + PDM(6CH) + I2S [0xD] SW0(4MDATA) + SW1(1MDATA) + PDM(8CH) [0xE] LPFLL Clock out [0xF] SW0(3MDATA) + SW1(2MDATA) [0x10] SW0(3MDATA) + SW1(2MDATA) + PDM(2CH) [0x11] SW0(3MDATA) + SW1(2MDATA) + PDM(8CH) [0x12] SW0(3MDATA) + SW1(2MDATA) + PDM(6CH) + I2S [0x13] 2I2S + 1 INTR + PDM(4CH) [0x14]

FCH Common Options (0x7005)

Submenus I3C/I2C Configuration Options ■ USB Configuration Options ■ Ac Power Loss Options ■ Uart Configuration Options ■ USB4 Configuration Options ■ SPI Configuration Options

I3C/I2C Configuration Options (0x7037)

Setting	Type	Options / Values / Range
I3C/I2C 0 Enable	Selection	Both Disabled [0x3] I3C Enabled [0x0] I2C Enabled [0x1] Auto [0xF] ■ default
I3C 0 Mode	Selection	I3C [0x0] I2C [0x1] Auto [0xF] ■ default
I3C/I2C 1 Enable	Selection	Both Disabled [0x3] I3C Enabled [0x0] I2C Enabled [0x1] Auto [0xF] ■ default
I3C 1 Mode	Selection	I3C [0x0] I2C [0x1] Auto [0xF] ■ default
I3C/I2C 2 Enable	Selection	Both Disabled [0x3] I3C Enabled [0x0] I2C Enabled [0x1] Auto [0xF] ■ default
I3C 2 Mode	Selection	I3C [0x0] I2C [0x1] Auto [0xF] ■ default
I3C/I2C 3 Enable	Selection	Both Disabled [0x3] I3C Enabled [0x0] I2C Enabled [0x1] Auto [0xF] ■ default
I3C 3 Mode	Selection	I3C [0x0] I2C [0x1] Auto [0xF] ■ default

USB Configuration Options (0x7038)

Submenus USB0 2.0 port enable ■ USB1 2.0 port enable ■ USB2 2.0 port enable ■ USB3 2.0 port enable ■ USB0 3.1 port enable ■ USB1 3.1 port enable ■ USB2 3.1 port enable ■ USB3 3.1 port enable

Setting	Type	Options / Values / Range
USB0 controller enable	Selection	Enabled [0x1] Disabled [0x0] Auto [0xF] ■ default
USB1 controller enable	Selection	Enabled [0x1] Disabled [0x0] Auto [0xF] ■ default

USB0 2.0 port enable (0x703D)

Setting	Type	Options / Values / Range
USB0 2.0 Port 0	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default
USB0 2.0 Port 1	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default
USB0 2.0 Port 2	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default
USB0 2.0 Port 3	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default
USB0 2.0 Port 4	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default

USB1 2.0 port enable (0x703E)

Setting	Type	Options / Values / Range
USB1 2.0 Port 0	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default

USB2 2.0 port enable (0x703F)

Setting	Type	Options / Values / Range
USB2 2.0 Port 0	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default

USB3 2.0 port enable (0x7040)

Setting	Type	Options / Values / Range
USB3 2.0 Port 0	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default

USB0 3.1 port enable (0x7041)

Setting	Type	Options / Values / Range
USB0 3.1 Port 0	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default
USB0 3.1 Port 1	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default

USB1 3.1 port enable (0x7042)

Setting	Type	Options / Values / Range
USB1 3.1 Port 0	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default

USB2 3.1 port enable (0x7043)

Setting	Type	Options / Values / Range
USB2 3.1 Port 0	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default

USB3 3.1 port enable (0x7044)

Setting	Type	Options / Values / Range
USB3 3.1 Port 0	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default

Ac Power Loss Options (0x7039)

Setting	Type	Options / Values / Range
Ac Loss Control	Selection	Always Off [0x0] ■ default Always On [0x1] Last State [0x3]

Uart Configuration Options (0x703A)

Setting	Type	Options / Values / Range
Uart 0 Enable	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default
Uart 0 Legacy Options	Selection	Disabled [0x0] 0x2E8 [0x1] 0x2F8 [0x2] 0x3E8 [0x3] 0x3F8 [0x4] Auto [0xF] ■ default
Uart 1 Enable (no HW FC)	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default
Uart 1 Legacy Options	Selection	Disabled [0x0] 0x2E8 [0x1] 0x2F8 [0x2] 0x3E8 [0x3] 0x3F8 [0x4] Auto [0xF] ■ default
Uart 2 Enable	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default

Setting	Type	Options / Values / Range
Uart 2 Legacy Options	Selection	Disabled [0x0] 0x2E8 [0x1] 0x2F8 [0x2] 0x3E8 [0x3] 0x3F8 [0x4] Auto [0xF] ■ default
Uart 3 Enable (no HW FC)	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default
Uart 3 Legacy Options	Selection	Disabled [0x0] 0x2E8 [0x1] 0x2F8 [0x2] 0x3E8 [0x3] 0x3F8 [0x4] Auto [0xF] ■ default
Uart 4 Enable	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default

USB4 Configuration Options (0x703B)

Submenus USB4 RT0 ■ USB4 RT1

Setting	Type	Options / Values / Range
USB4 pre-CM Enable	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default

USB4 RT0 (0x7045)

Setting	Type	Options / Values / Range
RT0 Router Enable	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default
RT0 PCIe Tunnel	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default

USB4 RT1 (0x7046)

Setting	Type	Options / Values / Range
RT1 Router Enable	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default
RT1 PCIe Tunnel	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default

SPI Configuration Options (0x703C)

Setting	Type	Options / Values / Range
HFP Enable	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default
HID Enable	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default
HID2 Enable	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default
HID Spi Read Mode	Selection	Quad IO 1-4-4 [0x5] ■ default Dual IO 1-2-2 [0x4] Fast Read 1-1-1 [0x7]
HID Spi Speed	Selection	(1)33HMz [0x1] ■ default (2)22NMz [0x2] (3)17.6Mhz [0x3] (5)800Khz [0x5] spi_spd6 [0x6] spi_spd7 [0x7]
HID Spi Speed6 N	Numeric	Min 0x4 · Max 0x3F · Default 0x4 ■ default
HID Spi Speed7 N	Numeric	Min 0x4 · Max 0x3F · Default 0x32 ■ default
HID2 Spi Read Mode	Selection	Quad IO 1-4-4 [0x5] ■ default Dual IO 1-2-2 [0x4] Fast Read 1-1-1 [0x7]
HID2 Spi Speed	Selection	(1)33HMz [0x1] ■ default (2)22NMz [0x2] (3)17.6Mhz [0x3] (5)800Khz [0x5] spi_spd6 [0x6] spi_spd7 [0x7]
HID2 Spi Speed6 N	Numeric	Min 0x4 · Max 0x3F · Default 0x4 ■ default
HID2 Spi Speed7 N	Numeric	Min 0x4 · Max 0x3F · Default 0x32 ■ default

SMU Common Options (0x7006)

Setting	Type	Options / Values / Range
System Configuration	Selection	15W [0xE] 28W [0xF] 30W [0x6] 35W [0x7] 45W [0x4] 54W [0x9] 65W [0xA] Auto [0xFF] 55W [0x1] 85W [0x2] 120W [0x3] ■ default 140W [0x5] ■ performance TUNE Platform cTDP profile. The compiled default is 120W; 140W is the highest profile this board exposes and the one to select for maximum sustained clocks. Lower profiles cap power for quiet or cool operation.

Setting	Type	Options / Values / Range
ECO Mode	Selection	Auto [0xF] ■ default Enabled [0x1] Disabled [0x0] ■ performance TUNE Disabled for maximum performance. ECO caps APU power for efficiency/quiet at the cost of sustained clocks.
SPL Control	Selection	Manual [0x1] Auto [0x0] ■ default ■ performance TUNE Manual to raise the Sustained Power Limit; higher SPL = more sustained all-core/iGPU clock within thermal limits.
Sustained Power Limit	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ default ■ performance TUNE Raise toward the GTR9 Pro ceiling (~140 W) for best sustained throughput; bounded by cooling and skin-temp limits.
PPT Control	Selection	Manual [0x1] Auto [0x0] ■ default ■ performance TUNE Manual to lift Fast/Slow PPT limits for more boost headroom (cooling permitting).
Fast PPT Limit	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ default ■ performance TUNE Higher Fast PPT allows larger short-term power spikes for snappier boost; keep within thermal headroom.
Slow PPT Limit	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ default ■ performance TUNE Higher Slow PPT raises the sustained power ceiling for long workloads.
Slow PPT Time Constant	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ default
STAPM Control	Selection	Manual [0x1] Auto [0x0] ■ default ■ performance TUNE Manual to relax STAPM (skin-temp-aware power) for more sustained performance; default protects chassis temperature.
System Temperature Tracking	Selection	Auto [0xF] ■ default 1 [0x1] 0 [0x0] ■ performance TUNE Disabling skin-temp tracking removes a sustained-power cap (more performance, hotter chassis); keep enabled for safe thermals.
STAPM Boost Override	Selection	Auto [0xF] ■ default 1 [0x1] 0 [0x0] ■ performance TUNE Set 1 to extend boost duration before STAPM clamps; raises chassis temperature. Auto follows the firmware policy.
STAPM Boost	Selection	Auto [0xF] ■ default 1 [0x1] 0 [0x0] ■ performance TUNE Higher STAPM boost prolongs high-power boost; bounded by chassis thermals.
Tskin Time Constant (STAPM)	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ default
Thermal Control	Selection	Manual [0x1] ■ default Auto [0x0] ■ performance TUNE Manual to raise TjMax-driven limits (only with strong cooling); default keeps the die at safe temps.

Setting	Type	Options / Values / Range
TjMax	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x5A ■ default ■ performance TUNE Default (typically 100 C). Lowering TjMax reduces boost; raising is not advisable. Leave at the AMD-validated junction limit.
TDC Control	Selection	Manual [0x1] Auto [0x0] ■ default ■ performance TUNE Manual to raise sustained current limits in step with PPT for all-core boost.
TDC_VDDCR_VDD	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ default ■ performance TUNE Core-rail sustained current cap — raise with PPT for sustained all-core clocks.
TDC_VDDCR_SOC	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ default
TDC_VDDCR_SR	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ default
EDC Control	Selection	Manual [0x1] Auto [0x0] ■ default ■ performance TUNE Manual to raise peak current limits for burst boost.
EDC_VDDCR_VDD	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ default ■ performance TUNE Core-rail peak current cap — raise for burst boost headroom.
EDC_VDDCR_SOC	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ default
EDC_VDDCR_SR	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ default
PSI3 Control	Selection	Manual [0x1] Auto [0x0] ■ default
PSI3_VDDCR_VDD	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ default
PROCHOT Control	Selection	Manual [0x1] Auto [0x0] ■ default ■ performance TUNE Leave Auto. PROCHOT is the thermal-safety throttle; disabling it risks the silicon and is not a sanctioned performance gain.
PROCHOT Deassertion Ramp Time	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ default
STT Control	Selection	Manual [0x1] Auto [0x0] ■ default ■ performance TUNE Skin-Temperature Tracking master — Manual to raise skin-temp limits for more sustained power (hotter chassis).
STT_PCB_SENSOR_COUNT	Numeric	Min 0x0 · Max 0xF · Default 0x0 ■ default
STT_MIN_POWER_LIMIT	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■ default
STT_M1	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■ default
STT_M2	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■ default
STT_M3	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■ default
STT_M4	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■ default
STT_M5	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■ default
STT_M6	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■ default

Setting	Type	Options / Values / Range
STT_C_APU	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■ default
STT_C_HS2	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■ default
STT_ALPHA_APU	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■ default
STT_ALPHA_HS2	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■ default
STT_SKIN_TEMPERATURE_LIMIT_APU	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■ default ■ performance TUNE Raising the APU skin-temp limit sustains boost longer at higher chassis temperature; default protects surface temp.
STT_SKIN_TEMPERATURE_LIMIT_HS2	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■ default ■ performance TUNE Raising the secondary skin-temp limit sustains more power; bounded by safe chassis temps.
STT_ERROR_COEFF	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■ default
STT_ERROR_RATE_COEFF	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■ default
Fan Control	Selection	Manual [0x1] Auto [0x0] ■ default ■ performance TUNE Use a graduated aggressive curve. More airflow = more thermal headroom = higher sustained boost (the documented GTR9 Pro tuning win).
Force PWM Control	Selection	Force [0x1] Unforce [0x0] ■ default ■ performance TUNE Force pins a fixed PWM duty for maximum cooling during benchmarks or sustained load, at the cost of noise. Unforce returns curve control.
Force PWM	Numeric	Min 0x0 · Max 0x64 · Default 0x0 ■ default ■ performance TUNE Higher fixed PWM raises airflow and sustained clocks at the cost of noise.
Fan Table Control	Selection	Manual [0x1] Auto [0x0] ■ default ■ performance TUNE Manual to define an aggressive temp/PWM table for better sustained thermals.
Low Temperature	Numeric	Min 0x0 · Max 0xFF · Default 0x0 ■ default ■ performance TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temp / higher PWM) to gain thermal headroom; trade against noise.
Medium Temperature	Numeric	Min 0x0 · Max 0xFF · Default 0x0 ■ default ■ performance TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temp / higher PWM) to gain thermal headroom; trade against noise.
High Temperature	Numeric	Min 0x0 · Max 0xFF · Default 0x0 ■ default ■ performance TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temp / higher PWM) to gain thermal headroom; trade against noise.
Critical Temperature	Numeric	Min 0x0 · Max 0xFF · Default 0x0 ■ default ■ performance TUNE Emergency thermal-shutdown threshold, not a fan-curve point. Leave at the firmware default - this is a safety limit.

Setting	Type	Options / Values / Range
Low Pwm	Numeric	Min 0x0 · Max 0x64 · Default 0x0 ■ default ■ performance TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temp / higher PWM) to gain thermal headroom; trade against noise.
Medium Pwm	Numeric	Min 0x0 · Max 0x65 · Default 0x0 ■ default ■ performance TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temp / higher PWM) to gain thermal headroom; trade against noise.
High Pwm	Numeric	Min 0x0 · Max 0x66 · Default 0x0 ■ default ■ performance TUNE Fan-curve point. For sustained performance, bias the curve toward more airflow (lower start temp / higher PWM) to gain thermal headroom; trade against noise.
Temperature Hysteresis	Numeric	Min 0x0 · Max 0xFF · Default 0x0 ■ default ■ performance TUNE Deadband before the fan steps back down. Larger values reduce fan hunting; smaller values shed heat sooner.
Pwm Frequency	Selection	Auto [0xF] ■ default 1 [0x1] 0 [0x0] ■ performance KEEP Auto. PWM carrier frequency. Changes fan noise character, not cooling capacity or curve shape.
Fan polarity	Selection	Auto [0xF] ■ default 1 [0x1] 0 [0x0] ■ performance KEEP Auto. PWM signal polarity for the fan header - a wiring compatibility setting, not a curve point. The wrong polarity inverts fan response.

SOC Miscellaneous Control (0x7007)

Submenus Pluton Options Firmware Anti-rollback (FAR) AIM-T Options Intrusion Detection SecureBio

Setting	Type	Options / Values / Range
Pluton Attribute Control	Selection	Disabled [0x0] Enabled [0x1] ■ default
Pluton fTPM Attribute Control	Selection	Disabled [0x0] Enabled [0x1] ■ default
Trusted Platform Module	Selection	Auto [0xFE] ■ default Disabled [0xFF] Enable dTPM [0x0] Enable ASP fTPM [0x1] Enable Pluton fTPM [0x2]
Pluton Support	Selection	Auto [0xF] ■ default Disabled [0x0] Enabled [0x1]
Pluton Security Processor	Selection	Auto [0xF] ■ default Disabled [0x0] Enabled [0x1]

Setting	Type	Options / Values / Range
Microsoft Security Levels	Selection	Customized [0xFF] ■ default dTPM Level 1 without Pluton Security Processor [0x1] dTPM Level 2 without Pluton Security Processor [0x2] dTPM Level 3 without Pluton Security Processor [0x3] dTPM Level 1 with Pluton Security Processor [0xB] dTPM Level 2 with Pluton Security Processor [0xC] dTPM Level 3 with Pluton Security Processor [0xD] Pluton fTPM Level 1 without Pluton Security Processor [0x15] Pluton fTPM Level 2 without Pluton Security Processor [0x16] Pluton fTPM Level 3 without Pluton Security Processor [0x17] Pluton fTPM Level 1 with Pluton Security Processor [0x1F] Pluton fTPM Level 2 with Pluton Security Processor [0x20] Pluton fTPM Level 3 with Pluton Security Processor [0x21] ASP fTPM Level 0 [0x28] ASP fTPM Level 1 without Pluton Security Processor [0x29] ASP fTPM Level 2 without Pluton Security Processor [0x2A] ASP fTPM Level 1 with Pluton Security Processor [0x33] ASP fTPM Level 2 with Pluton Security Processor [0x34]
Secured-core Auto enablement	Selection	Auto [0xFF] ■ default Enabled [0x1] Disabled [0x0]
DRTM Support	Selection	Auto [0xFF] ■ default Enabled [0x1] Disabled [0x0]
SMM Isolation Support	Selection	Auto [0xFF] ■ default Enabled [0x1] Disabled [0x0]
Pro Part attribute control	Selection	0 [0x0] ■ default 1 [0x1]
GFX workstation support	Selection	Auto [0xF] ■ default Enabled [0x1] Disabled [0x0]
ABL Console Out Control	Selection	Auto [0x2] ■ default Enable [0x1] Disable [0x0]
ABL Console Out Serial Port	Selection	Auto [0xFF] ■ default LPC UART [0x0] FCH UART0 [0x1] FCH UART1 [0x2] FCH UART2 [0x3] FCH UART3 [0x4]
ABL Console Out Boot Mode Select	Selection	ALL [0xFF] ■ default S3/S0i3 only [0x2] Normal boot only [0x1]
ABL PMU message Control	Selection	Auto [0xFF] ■ default Maximal debug messages [0x4] Detailed debug message [0x5] Coarse debug message [0xA] Stage completion [0xC8] Assertion message [0xC9] Firmware completion message only [0xFE]

Setting	Type	Options / Values / Range
PSP RPMC Switch	Selection	Auto [0xFF] ■ default Disabled [0x0] Enabled [0x1]
ABL Training Mode	Selection	default [0x0] ■ default serial [0x1] parallel [0x2]

Pluton Options (0x7056)

Setting	Type	Options / Values / Range
Pluton UART	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default
Pluton UART Port	Selection	UART 0 [0x0] UART 1 [0x1] UART 2 [0x2] UART 3 [0x3] UART 4 [0x4] UART 5 [0x5] UART 6 [0x6] UART 7 [0x7] Auto [0xF] ■ default
Pluton FIPS	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default
Pluton FIPS zeroization	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default

Firmware Anti-rollback (FAR) (0x7057)

Setting	Type	Options / Values / Range
FAR enforcement state	Selection	Enabled [0x1] Disabled [0x0] ■ default
SPL value in the CPU fuse	Numeric	Min 0x0 · Max 0xFF · Default 0x0 ■ default
SPL value in the SPL table	Numeric	Min 0x0 · Max 0xFF · Default 0x0 ■ default
Initial SPL value for FAR	Numeric	Min 0x0 · Max 0xFF · Default 0x0 ■ default
SPL update switch	Selection	Auto [0xFF] ■ default Enabled [0x1] Disabled [0x0]

AIM-T Options (0x7058)

Setting	Type	Options / Values / Range
AIM-T Support	Selection	AIM-T Disabled [0x0] AIM-T Enabled [0x1] Auto [0xF] ■ default

Setting	Type	Options / Values / Range
MPM attribute control	Selection	0 [0x0] ■ default 1 [0x1] 2 [0x2] 3 [0x3] 4 [0x4] 5 [0x5] 6 [0x6] 7 [0x7]
KVM for Wired Manageability	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default
Wireless Manageability	Selection	Disabled [0x0] Enabled [0x1] Auto [0xF] ■ default

Intrusion Detection (0x7059)

Setting	Type	Options / Values / Range
Intrusion Detection Control	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■ default
Intrusion Detection Enable	Selection	Auto [0xFF] ■ default Enabled [0x1] Disabled [0x0]
Log Intrusion Event	Selection	Enabled [0x1] Disabled [0x0] ■ default
Clear TPM	Selection	Enabled [0x1] Disabled [0x0] ■ default
Power Off System	Selection	Enabled [0x1] Disabled [0x0] ■ default

SecureBio (0x705A)

Setting	Type	Options / Values / Range
SecureBio Support	Selection	Auto [0xF] ■ default Enable [0x1] Disable [0x0]
SecureBio Camera Support	Selection	Disabled [0x0] ■ default XHCI Camera [0x1] MIPI Camera [0x2] Auto [0xF]

AMD PBS (Platform BIOS Settings)

14 forms · 202 settings

AMD PBS Option (0xB)

Submenus AMD Firmware Version ■ PCI Express Configurations ■ Power Saving Configurations ■ Graphics Configurations ■ Display Configurations ■ USB/Thunderbolt Configurations ■ USB TypeC Security Mode Configurations ■ Audio Configurations ■ I2C Configurations ■ Thermal Configurations ■ MP2 Configurations ■ EC/PD Configurations ■ Debug Configurations

Setting	Type	Options / Values / Range
Enable SLT check	Selection	Disabled [0x0] Enabled [0x1] ■ default Force SLT check halt [0x2]
Above 4GB MMIO Limit	Selection	35bit (32GB) [0x23] 36bit (64GB) [0x24] 37bit (128GB) [0x25] 38bit (256GB) [0x26] 39bit (512GB) [0x27] 40bit (1TB) [0x28] ■ default 41bit (2TB) [0x29] 42bit (4TB) [0x2A] 43bit (8TB) [0x2B] 44bit (16TB) [0x2C] 45bit (32TB) [0x2D] 46bit (64TB) [0x2E] 47bit (128TB) [0x2F] 48bit (256TB) [0x30] ■ performance KEEP 40bit (1TB). This sizes the above-4G MMIO aperture; it is not the Above-4G Decoding enable. The default already covers the 96 GB iGPU carve-out plus every device BAR on this board.
Wireless LAN Recovery	Selection	Disabled [0x0] ■ default Enabled [0x1] Dummy reset [0x2]
Wireless LAN Country Code	Selection	Disabled [0x0] ■ default Country Code USA [0x1] Country Code China [0x2]
Bluetooth PLDR	Selection	Disabled [0x0] ■ default Enabled [0x1] Dummy reset [0x2]
Wireless Button	Selection	Disabled [0x0] ■ default HID Based [0x3]
Processor Aggregator Device	Selection	Disabled [0x0] ■ default Enabled [0x1]
Core Count Control	Selection	Disabled [0x0] ■ default Enabled [0x1] ■ performance KEEP Leave all cores enabled. Reducing core count lowers multi-thread performance; only down-core for special validation.
Core Count SW_SCI_GPE_ID	Numeric	Min 0x0 · Max 0x1F · Default 0x19 ■ default
APIC Software Enable	Selection	Disabled [0x0] Enabled [0x1] ■ default

Setting	Type	Options / Values / Range
Dynamic P3T limit	Selection	Disabled [0x0] Enable for DC-only case (include fake DC) [0x1] ■ default Enabled [0x2] ■ performance TUNE Dynamic platform power-tracking limit. Disabling pins P3T instead of letting firmware adapt it; leave Enabled unless you set power limits by hand.
Dynamic LID	Selection	Disabled [0x0] ■ default Enabled [0x1] ■ performance KEEP Lid-switch state handling. This mini-PC has no lid, so the setting has no runtime effect. Leave default.
HDMI 3.0G Tx SLEW	Selection	Disabled [0x0] ■ default Enabled [0x1]
HDMI 3.0G Tx Slew Control Value	Numeric	Min 0x0 · Max 0xFF · Default 0x14 ■ default
Control Method Button Support	Selection	Legacy Power Button (PNP0C0C) [0x0] ■ default Generic Button (ACPI0011) [0x1]

AMD Firmware Version (0xC)

Information fields AGESA Version, PSP BootLoader Version, PSP SecureOS Version, ABL Version, APCB Version, APOB Version, DF Rib Version, Ucode Patch Version, S3 Image ID, SMU FW Version, ZSC FW Version, MPIO FW Version, MP2 FW Version, XHCI FW Version, IMU FW Version, VBIOS FW Version, GOP Driver Version, EC FW Version, EDKII Version, PD0 FW Version, PD0 Custom, PD0 SKU, PD0 Mini Type, Re-timer, PD1 FW Version, PD1 Custom, PD1 SKU, PD1 Mini Type, AIM-T(MPM) Part1 FW Version, AIM-T(MPM) Part2 FW Version

PCI Express Configurations (0xD)

Setting	Type	Options / Values / Range
Pcie Dxio Timing ControlEnable	Selection	Auto [0xF] ■ default Enabled [0x1] Disabled [0x0] ■ performance TUNE Exposes DXIO link-training timing overrides. Leave Auto; these help marginal links train, they do not raise steady-state PCIe performance.
PCIE Link Receiver Detection Polling	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ default ■ performance TUNE Link-training receiver-detection timing. Leave default; it affects enumeration of marginal links, not link speed or latency.
PCIE Link L0 Polling	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ default ■ performance TUNE Link-training poll interval, not a link power state. Leave default; this governs enumeration reliability rather than throughput.
Enable power sequence of specific WWAN	Selection	Disabled [0x0] ■ default Power Sequence — Fibocom [0x1] Power Sequence — Quectel RM520 [0x2] Power Sequence — Quectel EM160 [0x3]
PCIE x4 DT Slot Power Enable	Selection	Disabled [0x0] Enabled [0x1] ■ default
M.2 SSD1 Power Enable	Selection	Disabled [0x0] Enabled [0x1] ■ default

Setting	Type	Options / Values / Range
WLAN Power Enable	Selection	Disabled [0x0] x1 [0x1] ■ default x2 [0x2]
WWAN Power Enable	Selection	Disabled [0x0] x1 [0x1] ■ default x2 [0x2]
SD Card Reader Power Enable	Selection	Disabled [0x0] Enabled [0x1] ■ default
Initial SD7 L1 Substates at OS runtime	Selection	Disabled [0x0] ■ default Enabled [0x1]
GbE Power Enable	Selection	Disabled [0x0] Enabled [0x1] ■ default
M.2 SSD0 Power Enable	Selection	Disabled [0x0] Enabled [0x1] ■ default
NVMe RAID mode	Selection	Disabled [0x0] ■ default Enabled [0x1]
Save Restore DXIO Feature	Selection	Auto [0xF] ■ default Disabled [0x0] Enabled [0x1]

Power Saving Configurations (0xE)

Setting	Type	Options / Values / Range
NVME D3Cold	Selection	Disabled [0x0] ■ default Enabled [0x1] ■ performance TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal perf); Enabled saves idle power. Default is fine for desktop use.
M.2 NVME Tpvperl	Numeric	Min 0x32 · Max 0x3E8 · Default 0x32 ■ default
M.2 NVME Trst-cfg	Numeric	Min 0x64 · Max 0x3E8 · Default 0x64 ■ default
WLAN D3Cold	Selection	Disabled [0x0] ■ default Enabled [0x1] ■ performance TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal perf); Enabled saves idle power. Default is fine for desktop use.
LOM D3Cold	Selection	Disabled [0x0] ■ default Enabled [0x1] ■ performance TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal perf); Enabled saves idle power. Default is fine for desktop use.
SD CardReader D3Cold	Selection	Disabled [0x0] ■ default Enabled [0x1] ■ performance TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal perf); Enabled saves idle power. Default is fine for desktop use.
USB4 D3 Eanble	Selection	Disabled [0x0] Enabled [0x1] ■ default

Setting	Type	Options / Values / Range
Internal PCIe GPP 0 D3	Selection	Disabled [0x0] ■ default Enabled [0x4] ■ performance TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal perf); Enabled saves idle power. Default is fine for desktop use.
SOC GPU D3	Selection	Disabled [0x0] ■ default Enabled [0x4] ■ performance TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal perf); Enabled saves idle power. Default is fine for desktop use.
SOC HD Audio D3	Selection	Disabled [0x0] Enabled [0x4] ■ default ■ performance TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal perf); Enabled saves idle power. Default is fine for desktop use.
SOC USB3.1 D3	Selection	Disabled [0x0] Enabled [0x4] ■ default
SOC ACP D3	Selection	Disabled [0x0] Enabled [0x4] ■ default ■ performance TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal perf); Enabled saves idle power. Default is fine for desktop use.
SOC Azalia D3	Selection	Disabled [0x0] Enabled [0x4] ■ default ■ performance TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal perf); Enabled saves idle power. Default is fine for desktop use.
Internal PCIe GPP 2 D3	Selection	Disabled [0x0] ■ default Enabled [0x4] ■ performance TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal perf); Enabled saves idle power. Default is fine for desktop use.
SOC USB3.1 for USB4 D3	Selection	Disabled [0x0] Enabled [0x4] ■ default
SOC USB4 D3	Selection	Disabled [0x0] Enabled [0x4] ■ default
Internal USB4 PCIe Tunneling D3	Selection	Disabled [0x0] Enabled [0x4] ■ default
SOC USB4 PCIe Endpoint D3	Selection	Disabled [0x0] Enabled [0x4] ■ default
Keep Wlan Power In S3/S4 state	Selection	Disabled [0x0] ■ default Enable Wlan Power In S3 And S4 [0x3]
Unused GPP Clocks Off	Selection	Disabled [0x0] ■ default Enabled [0x1] ■ performance TUNE Gates the reference clock for unpopulated PCIe GPP ports. Disabled keeps all clocks running; Enabled saves idle power with no effect on active links.

Setting	Type	Options / Values / Range
Clock PM: CLK_REQ0	Selection	Disabled [0x0] Enabled [0x1] ■ default ■ performance TUNE PCIe link power management. Disabled = full link speed and lowest latency (best performance, higher idle power); Enabled (L1) = power saving with wake latency. Performance: Disabled.
Clock PM: CLK_REQ1	Selection	Disabled [0x0] Enabled [0x1] ■ default ■ performance TUNE PCIe link power management. Disabled = full link speed and lowest latency (best performance, higher idle power); Enabled (L1) = power saving with wake latency. Performance: Disabled.
Clock PM: CLK_REQ2	Selection	Disabled [0x0] Enabled [0x1] ■ default ■ performance TUNE PCIe link power management. Disabled = full link speed and lowest latency (best performance, higher idle power); Enabled (L1) = power saving with wake latency. Performance: Disabled.
Clock PM: CLK_REQ3	Selection	Disabled [0x0] Enabled [0x1] ■ default ■ performance TUNE PCIe link power management. Disabled = full link speed and lowest latency (best performance, higher idle power); Enabled (L1) = power saving with wake latency. Performance: Disabled.
Clock PM: CLK_REQ4	Selection	Disabled [0x0] Enabled [0x1] ■ default ■ performance TUNE PCIe link power management. Disabled = full link speed and lowest latency (best performance, higher idle power); Enabled (L1) = power saving with wake latency. Performance: Disabled.
Clock PM: CLK_REQ5	Selection	Disabled [0x0] Enabled [0x1] ■ default ■ performance TUNE PCIe link power management. Disabled = full link speed and lowest latency (best performance, higher idle power); Enabled (L1) = power saving with wake latency. Performance: Disabled.
Clock PM: CLK_REQ6	Selection	Disabled [0x0] Enabled [0x1] ■ default ■ performance TUNE PCIe link power management. Disabled = full link speed and lowest latency (best performance, higher idle power); Enabled (L1) = power saving with wake latency. Performance: Disabled.
PCIe x4 Slot D3 Cold	Selection	Disabled [0x0] ■ default Renesas XHCI [0x1] Realtek SD Card Reader [0x2] ■ performance TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal perf); Enabled saves idle power. Default is fine for desktop use.

USB/Thunderbolt Configurations (0x17)

Setting	Type	Options / Values / Range
PD USB4 Control Enable/Disable	Selection	Disabled [0x0] Enabled [0x1] ■ default
USB4 Bus Reserved	Numeric	Min 0x0 · Max 0xFF · Default 0x60 ■ default
USB4 IO Reserved (KB)	Numeric	Min 0x0 · Max 0xFF · Default 0x10 ■ default

Setting	Type	Options / Values / Range
USB4 Non-Prefetch Memory Reserved	Numeric	Min 0x0 · Max 0xFFFF · Default 0x180 ■ default
USB4 non-Prefetch MMIO align (0 ~ 65534 MB)	Numeric	Min 0x0 · Max 0xFFFF · Default 0x40 ■ default
USB4 Prefetch Memory Reserved	Selection	2MB [0x2] 4MB [0x4] 8MB [0x8] 16MB [0x10] 32MB [0x20] 64MB [0x40] 128MB [0x80] 256MB [0x100] 512MB [0x200] 1GB [0x400] 2GB [0x800] 4GB [0x1000] 8GB [0x2000] 16GB [0x4000] 32GB [0x8000] 64GB [0x10000] 128GB [0x20000] ■ default
USB4 Prefetch MMIO align	Selection	2MB [0x2] 4MB [0x4] 8MB [0x8] 16MB [0x10] 32MB [0x20] 64MB [0x40] 128MB [0x80] 256MB [0x100] 512MB [0x200] 1GB [0x400] 2GB [0x800] 4GB [0x1000] 8GB [0x2000] 16GB [0x4000] 32GB [0x8000] ■ default 64GB [0x10000]
USB Camera Enable	Selection	Disabled [0x0] Enabled [0x1] ■ default
USB Fingerprint or GBE MUX	Selection	Fingerprint [0x0] GBE [0x1] ■ default
USB Fingerprint Enable	Selection	Disabled [0x0] ■ default Enabled [0x1]
UCSI Support	Selection	Disabled [0x0] ■ default Enabled [0x1]
UCSI tunnel location	Selection	UCSI tunnel at EC RAM [0x0] UCSI tunnel at MMIO 0xFEEC2000 [0x1] ■ default
USBC Port Hardware Disable Support	Selection	Disabled [0x0] ■ default Enabled [0x1]
Reconfig Rebalance Resources by OS	Selection	Disabled [0x0] ■ default Enabled [0x1]

Setting	Type	Options / Values / Range
PD Thunderbolt3 Alt Mode	Selection	Enabled [0x0] ■ default Disabled [0x1]
USB4 ACPI _DEP Support	Selection	Disabled [0x0] ■ default Enabled [0x1]
USB4 LTR Snoop/NonSnoop Scale	Numeric	Min 0x0 · Max 0x7 · Step 0x1 · Default 0x2 ■ default
USB4 LTR Snoop/NonSnoop Value	Numeric	Min 0x1 · Max 0x3FF · Step 0x1 · Default 0xC8 ■ default
Modify _UPC/_PLD settings for USB3.2-A vertical port (Requires board rework)	Selection	Disabled [0x0] ■ default Enabled [0x1]

USB TypeC Security Mode Configurations (0x18)

Setting	Type	Options / Values / Range
USB Type-C Security	Selection	Auto [0xF] ■ default Mode 0 [0x0] Mode 1 [0x1] Mode 2 [0x2]

Graphics Configurations (0xF)

Setting	Type	Options / Values / Range
EVAL Slot Power Enable	Selection	Disabled [0x0] Enabled [0x1] ■ default
EVAL CARD T-Diode Routing Select	Selection	APU SMBUS1 [0x0] EC [0x1] ■ default
Special Display Features	Selection	Disabled [0x0] HybridGraphics [0x4] ■ default
D3Cold Support	Selection	Disabled [0x0] ■ default Enabled [0x1] Dummy D3Cold [0x2] ■ performance TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal perf); Enabled saves idle power. Default is fine for desktop use.
Discrete GPU Hotplug Mode	Selection	Basic Mode [0x0] Enhanced Mode [0x1] ■ default Non-Hotplug Mode [0x2]
Discrete GPU D3Cold HPD Support	Selection	Disabled [0x0] ■ default Enabled [0x1] ■ performance TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal perf); Enabled saves idle power. Default is fine for desktop use.
PME Turn Off Support	Selection	Disabled [0x0] Enabled [0x1] ■ default ■ performance TUNE PCIe PME_Turn_Off handshake during power-state transitions. Affects suspend/resume correctness, not throughput. Leave default.

Setting	Type	Options / Values / Range
D3Cold Force Gen1	Selection	Disabled [0x0] ■ default Enabled [0x1] ■ performance TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal perf); Enabled saves idle power. Default is fine for desktop use.
SLOT_PWR-PWREN Timing(ms)	Numeric	Min 0x0 · Max 0x32 · Step 0x1 · Default 0x1 ■ default
PWREN-RST Timing (ms)	Numeric	Min 0x0 · Max 0x96 · Step 0x1 · Default 0x50 ■ default
PERST-WAKEL23 Timing (ms)	Numeric	Min 0x1 · Max 0x1E · Default 0xA ■ default
DLACT-CFGACC Timing (ms)	Numeric	Min 0x1 · Max 0x64 · Default 0x64 ■ default
NVIDIA DGPU Power Enable	Selection	Disabled [0x0] ■ default Enabled [0x1]
Discrete GPU _DSM Function A	Selection	Disabled [0x0] ■ default Enabled [0x1]
Discrete GPU _DSM Function B	Selection	Disabled [0x0] ■ default Enabled [0x1]
Non-Eval Discrete GPU Support	Selection	Disabled [0x0] ■ default Enabled [0x1]
Discrete GPU HPD Circuitry	Selection	OR Circuitry [0x0] ■ default Pulse Circuitry [0x1]
Discrete GPU's USB Port	Selection	Keep Default Setting [0x0] ■ default Disabled [0x1]
Discrete GPU's SSID/SVID	Selection	Keep Default Setting [0x0] Program by Vendor [0x1] ■ default
Discrete GPU's VGA SSID/SVID	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ default
Discrete GPU's AUDIO SSID/SVID	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ default
Discrete GPU BOMACO Support	Selection	Disabled [0x0] ■ default Enabled [0x1]
MACO-PWR Timing (us)	Numeric	Min 0xA · Max 0xFF · Step 0x1 · Default 0xA ■ default
RST-MACO Timing (ms)	Numeric	Min 0x0 · Max 0xA · Step 0x1 · Default 0x0 ■ default
ATCS Function 9 Support	Selection	Disabled [0x0] ■ default Enabled [0x1]
AC Maximum Performance Limit	Numeric	Min 0x0 · Max 0x64 · Default 0x0 ■ default ■ performance TUNE AC/DC power-profile cap. On this mains-powered mini-PC the AC profile applies — set the AC limits to the highest/Maximum-Performance tier; DC limits are irrelevant.
AC Better Performance Limit	Numeric	Min 0x0 · Max 0x64 · Default 0x0 ■ default ■ performance TUNE AC/DC power-profile cap. On this mains-powered mini-PC the AC profile applies — set the AC limits to the highest/Maximum-Performance tier; DC limits are irrelevant.
AC Better Battery Limit	Numeric	Min 0x0 · Max 0x64 · Default 0x0 ■ default ■ performance TUNE AC/DC power-profile cap. On this mains-powered mini-PC the AC profile applies — set the AC limits to the highest/Maximum-Performance tier; DC limits are irrelevant.

Setting	Type	Options / Values / Range
DC Maximum Performance Limit	Numeric	Min 0x0 · Max 0x64 · Default 0x0 ■ default ■ performance TUNE AC/DC power-profile cap. On this mains-powered mini-PC the AC profile applies — set the AC limits to the highest/Maximum-Performance tier; DC limits are irrelevant.
DC Better Performance Limit	Numeric	Min 0x0 · Max 0x64 · Default 0x0 ■ default ■ performance TUNE AC/DC power-profile cap. On this mains-powered mini-PC the AC profile applies — set the AC limits to the highest/Maximum-Performance tier; DC limits are irrelevant.
DC Better Battery Limit	Numeric	Min 0x0 · Max 0x64 · Default 0x0 ■ default ■ performance TUNE AC/DC power-profile cap. On this mains-powered mini-PC the AC profile applies — set the AC limits to the highest/Maximum-Performance tier; DC limits are irrelevant.
DC Battery Saver Limit	Numeric	Min 0x0 · Max 0x64 · Default 0x0 ■ default ■ performance TUNE AC/DC power-profile cap. On this mains-powered mini-PC the AC profile applies — set the AC limits to the highest/Maximum-Performance tier; DC limits are irrelevant.
BLINK LED	Selection	Disabled [0x0] Enabled [0x1] ■ default GPIO 11 Output Low [0x2] GPIO 11 Output High [0x3]
ATIF Notify Command Code	Selection	Notify VGA 0x81 [0x0] ■ default Notify VGA 0xD0 [0xD0] Notify VGA 0xD1 [0xD1] Notify VGA 0xD2 [0xD2] Notify VGA 0xD3 [0xD3] Notify VGA 0xD4 [0xD4] Notify VGA 0xD5 [0xD5] Notify VGA 0xD6 [0xD6] Notify VGA 0xD7 [0xD7] Notify VGA 0xD8 [0xD8] Notify VGA 0xD9 [0xD9]
ATIF Function 21 Support	Selection	Disabled [0x0] ■ default Enabled [0x1]
External Graphics Port	Selection	Disabled [0x0] ■ default Enabled [0x1]
Hide XConnect GUI	Selection	Disabled [0x0] ■ default Enabled [0x1]
Run Time PM and D3	Selection	Disabled [0x0] ■ default Enabled [0x1] ■ performance TUNE Deep device-idle (D3Cold) power gating. Disabling lowers wake latency for that device (marginal perf); Enabled saves idle power. Default is fine for desktop use.
Support ATIF ATPX	Selection	Disabled [0x0] ■ default Enabled [0x1]
ATIF Function 22 Support	Selection	Disabled [0x0] ■ default Undefined [0x1] Integrated Graphics [0x2] Discrete Graphics [0x3]

Setting	Type	Options / Values / Range
GPU Package Power Limit Value	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ default ■ performance TUNE Within the shared APU budget, a higher GPU power slice favors graphics-bound work; it competes with CPU power on this single-budget APU.
ATIF Function 23 Support	Selection	Disabled [0x0] ■ default Enabled [0x1]
Vari-Bright Maximum Performance	Numeric	Min 0x0 · Max 0x4 · Default 0x0 ■ default
Vari-Bright Better Performance	Numeric	Min 0x0 · Max 0x4 · Default 0x1 ■ default
Vari-Bright Better Battery	Numeric	Min 0x0 · Max 0x4 · Default 0x3 ■ default
Vari-Bright Battery Saver	Numeric	Min 0x0 · Max 0x4 · Default 0x3 ■ default
ATIF Function 24 Support	Selection	Disabled [0x0] Enabled [0x1] ■ default
ISP Device number	Numeric	Min 0x0 · Max 0x64 · Step 0x1 · Default 0x0 ■ default
Primary Video Adaptor	Selection	Int Graphics (IGD) [0x1] ■ default Ext Graphics (PEG) [0x2]
Smart Mux Acpi Control	Selection	Disabled [0x0] ■ default Enabled without _DEP [0x1] Enabled [0x3]
Display Panel Multiplexer	Numeric	Min 0x0 · Max 0x7 · Default 0x0 ■ default
Smart Mux _HID Selection	Selection	SMUX1234 [0x0] ■ default MSFT0007 [0x1]
Smart Mux MDM Support Level	Selection	No Support [0x0] ■ default Development Support [0x1] Experimental Support [0x2] Full Support [0x3]
Smart Mux First Connected GPU	Selection	Integrated Graphics [0x0] ■ default Discrete Graphics [0x1]
Smart Mux ACPI Method Location	Selection	Under Mux [0x1] Under dGPU [0x2] ■ default
Verify PEI GOP	Selection	Disabled [0x0] ■ default Enabled [0x1]
Ultra mode support	Selection	Disabled [0x0] ■ default Enabled [0x1]

Display Configurations (0x10)

Setting	Type	Options / Values / Range
DP0	Selection	Default [0x0] ■ default EDP display [0x1] DP display [0x2] HDMI display [0x3]
Adjust DP Caps	Selection	Soc Default [0x0] ■ default Cap override [0x1]
DP0 Display Caps	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0xFFFFFFFF ■ default

Setting	Type	Options / Values / Range
DP1	Selection	Default [0x0] ■ default EDP display [0x1] DP display [0x2] HDMI display [0x3]
DP1 Display Caps	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0xFFFFFFFF ■ default
DP2	Selection	Default [0x0] ■ default EDP display [0x1] HDMI display [0x3] DP with TypeC display [0x4] DP without TypeC display [0x5]
DP2 Display Caps	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0xFFFFFFFF ■ default
DP3	Selection	Default [0x0] ■ default EDP display [0x1] HDMI display [0x3] DP with TypeC display [0x4] DP without TypeC display [0x5]
DP3 Display Caps	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0xFFFFFFFF ■ default
DP4	Selection	Default [0x0] ■ default EDP display [0x1] HDMI display [0x3] DP with TypeC display [0x4] DP without TypeC display [0x5]
DP4 Display Caps	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0xFFFFFFFF ■ default
EDP priority	Selection	EDP0 [0x0] ■ default EDP1 [0x1]

Audio Configurations (0x11)

Setting	Type	Options / Values / Range
Wake On Voice	Selection	Disabled [0x0] ■ default Enabled [0x1]
ACP Power Gating	Selection	Disabled [0x0] Enabled [0x1] ■ default ■ performance KEEP Enabled (Auto). Power-gates the audio co-processor when idle — frees power budget, no downside.
Verb Table Select	Selection	ALC245 Crb Default: SVID/SSID: 1022/D670 [0x0] ■ default ALC245_05112020 : SVID/SSID: 10EC/1208 [0x1] ALC256_06082020 : SVID/SSID: 10EC/1208 [0x2] ALC623 : SVID/SSID: 10EC/1208 [0x3] SN614X : SVID/SSID: 1022/D595 [0x4] CX11970EVK : SVID/SSID: 14F1/0101 [0x5]
USB Sideband Audio	Selection	Disabled [0x0] ■ default Enabled [0x1]

I2C Configurations (0x12)

Setting	Type	Options / Values / Range
Touch Panel Support	Selection	Under I2C 0 Bus [0x0] Under I2C 1 Bus [0x1] Under I2C 2 Bus [0x2] Under I2C 3 Bus [0x3] ELAN TSP Under I2C 0 Bus [0x10] ELAN TSP Under I2C 1 Bus [0x11] ELAN TSP Under I2C 2 Bus [0x12] ELAN TSP Under I2C 3 Bus [0x13] Under USB 6 Bus [0x4] Disabled [0x5] ■ default
Touch Pad Support	Selection	Under I2C 0 Bus [0x0] Under I2C 1 Bus [0x1] Under I2C 2 Bus [0x2] Under I2C 3 Bus [0x3] Disabled [0x4] ■ default
Touchpad Slave address	Numeric	Min 0x0 · Max 0xFF · Default 0x2C ■ default
Touchpad HID Descriptor address	Numeric	Min 0x0 · Max 0xFF · Default 0x20 ■ default
Nfc Support	Selection	Under I2C 0 Bus [0x0] Under I2C 1 Bus [0x1] Under I2C 2 Bus [0x2] Under I2C 3 Bus [0x3] Disabled [0x4] ■ default
MITT/WITT Selection	Selection	MITT Only [0x0] WITT Only [0x1] Both disable [0x2] ■ default
I2C MUX	Selection	Disabled [0x0] Enabled [0x1] ■ default
SMBUS0 Buffer	Selection	Disabled [0x0] Enabled [0x1] ■ default
SMBUS1 Buffer	Selection	Disabled [0x0] Enabled [0x1] ■ default

Thermal Configurations (0x13)

Setting	Type	Options / Values / Range
APU PROCHOT# setting	Selection	Disable APU_PROCHOT# [0x0] Enable APU_PROCHOT# in pure-DC case [0x1] Enable APU_PROCHOT# in pure-AC case [0x2] Enable APU_PROCHOT# in either pure-DC or pure-AC case (not AC+DC) [0x3] ■ default
AMD DPTC interface	Selection	Disabled [0x0] ■ default Enabled [0x1]
STT sensor reporting	Selection	Disabled [0x0] ■ default Report onboard sensors [0x1] Report onboard + eval card sensors [0x2]

MP2 Configurations (0x14)

Setting	Type	Options / Values / Range
Power Sensors Routing Select	Selection	WALLE lite PDT [0x0] ■ default WALLE lite PM log [0x1]
MP2 FW Selection	Selection	MP2_SFH [0x0] ■ default MP2_WalleLite [0x1]
Turn off Xtal (S3/S5)	Selection	Disabled [0x0] Enabled [0x1] ■ default
Sensor Fusion User Mode Driver	Selection	Disabled [0x0] Enabled [0x1] ■ default
Reserved Memory For MP2	Selection	Disabled [0xFF] 128 KB [0x0] ■ default 256 KB [0x1] 512 KB [0x2] 1 MB [0x3] 2 MB [0x4] 4 MB [0x5] 8 MB [0x6] 16 MB [0x7] 32 MB [0x8] 64 MB [0x9]
MP2 Privilege Mode	Selection	Disabled [0x0] ■ default Enabled [0x1]
Accelerator Sensor	Selection	Disabled [0x0] ■ default Enabled [0x1]
Magnet Sensor	Selection	Disabled [0x0] ■ default Enabled [0x1]
SRA Sensor	Selection	Disabled [0x0] ■ default Enabled [0x1]
Light Sensor	Selection	Disabled [0x0] ■ default Enabled [0x1]
Proximity Sensor	Selection	Disabled [0x0] ■ default Enabled [0x1]
Wake On Human Presence detection WA	Selection	Disabled [0x0] ■ default Enabled [0x1]
Gyroscope Sensor	Selection	Disabled [0x0] ■ default Enabled [0x1]
HPD-Lite Sensor	Selection	Disabled [0x0] ■ default Enabled [0x1]

EC/PD Configurations (0x15)

Setting	Type	Options / Values / Range
Charger mode BYPASS	Selection	Disabled [0x0] ■ default Enabled [0x1]
KBC Support	Selection	Disabled [0x0] Enabled [0x1] ■ default

Setting	Type	Options / Values / Range
AcDcSwitch	Selection	Disabled [0x0] ■ default Enabled [0x1]
AmdDcTimer	Numeric	Min 0x0 · Max 0xFF · Default 0x0 ■ default ■ performance TUNE DC-transition debounce timer, not a power cap. Not exercised on a mains-powered mini-PC; leave default.
AmdAcTimer	Numeric	Min 0x0 · Max 0xFF · Default 0x0 ■ default ■ performance TUNE AC-transition debounce timer, not a power cap. Only the AC path is exercised on a mains-powered mini-PC; leave default.
Fake DC Level	Numeric	Min 0x0 · Max 0x64 · Default 0x5E ■ default ■ performance TUNE Emulated battery level for validating the DC power path. Leave default; it does not cap performance while on AC.
VDD adjust for BoardDesign	Selection	Disabled [0x0] ■ default Enabled [0x1]
VDDIO_MEM_S3 voltage	Selection	voltage (1.05V) [0x0] ■ default voltage (+10mv) [0x1] voltage (+20mv) [0x2] voltage (+30mv) [0x3] voltage (+40mv) [0x4] voltage (+50mv) [0x5] voltage (+60mv) [0x6] voltage (+70mv) [0x7] voltage (+80mv) [0x8] voltage (+90mv) [0x9] voltage (+100mv) [0xA] voltage (-10mv) [0xB] voltage (-20mv) [0xC] voltage (-30mv) [0xD] voltage (-40mv) [0xE] voltage (-50mv) [0xF] voltage (-60mv) [0x10] voltage (-70mv) [0x11] voltage (-80mv) [0x12] voltage (-90mv) [0x13] voltage (-100mv) [0x14]

Setting	Type	Options / Values / Range
VDD MISC S5 voltage	Selection	voltage (0.75V) [0x0] ■ default voltage (+10mv) [0x1] voltage (+20mv) [0x2] voltage (+30mv) [0x3] voltage (+40mv) [0x4] voltage (+50mv) [0x5] voltage (+60mv) [0x6] voltage (+70mv) [0x7] voltage (+80mv) [0x8] voltage (+90mv) [0x9] voltage (+100mv) [0xA] voltage (-10mv) [0xB] voltage (-20mv) [0xC] voltage (-30mv) [0xD] voltage (-40mv) [0xE] voltage (-50mv) [0xF] voltage (-60mv) [0x10] voltage (-70mv) [0x11] voltage (-80mv) [0x12] voltage (-90mv) [0x13] voltage (-100mv) [0x14]
VDD_MISC voltage	Selection	voltage (0.75V) [0x0] ■ default voltage (+10mv) [0x1] voltage (+20mv) [0x2] voltage (+30mv) [0x3] voltage (+40mv) [0x4] voltage (+50mv) [0x5] voltage (+60mv) [0x6] voltage (+70mv) [0x7] voltage (+80mv) [0x8] voltage (+90mv) [0x9] voltage (+100mv) [0xA] voltage (-10mv) [0xB] voltage (-20mv) [0xC] voltage (-30mv) [0xD] voltage (-40mv) [0xE] voltage (-50mv) [0xF] voltage (-60mv) [0x10] voltage (-70mv) [0x11] voltage (-80mv) [0x12] voltage (-90mv) [0x13] voltage (-100mv) [0x14]

Setting	Type	Options / Values / Range
VDD11 voltage	Selection	voltage (1.1V) [0x0] ■ default voltage (+10mv) [0x1] voltage (+20mv) [0x2] voltage (+30mv) [0x3] voltage (+40mv) [0x4] voltage (+50mv) [0x5] voltage (+60mv) [0x6] voltage (+70mv) [0x7] voltage (+80mv) [0x8] voltage (+90mv) [0x9] voltage (+100mv) [0xA] voltage (-10mv) [0xB] voltage (-20mv) [0xC] voltage (-30mv) [0xD] voltage (-40mv) [0xE] voltage (-50mv) [0xF] voltage (-60mv) [0x10] voltage (-70mv) [0x11] voltage (-80mv) [0x12] voltage (-90mv) [0x13] voltage (-100mv) [0x14]
1V8 ALW voltage	Selection	voltage(1.8V) [0x0] ■ default voltage (+10mv) [0x1] voltage (+20mv) [0x2] voltage (+30mv) [0x3] voltage (+40mv) [0x4] voltage (+50mv) [0x5] voltage (+60mv) [0x6] voltage (+70mv) [0x7] voltage (+80mv) [0x8] voltage (+90mv) [0x9] voltage (+100mv) [0xA] voltage (-10mv) [0xB] voltage (-20mv) [0xC] voltage (-30mv) [0xD] voltage (-40mv) [0xE] voltage (-50mv) [0xF] voltage (-60mv) [0x10] voltage (-70mv) [0x11] voltage (-80mv) [0x12] voltage (-90mv) [0x13] voltage (-100mv) [0x14]

Setting	Type	Options / Values / Range
VDD2L voltage	Selection	voltage(0.9V) [0x0] ■ default voltage (+10mv) [0x1] voltage (+20mv) [0x2] voltage (+30mv) [0x3] voltage (+40mv) [0x4] voltage (+50mv) [0x5] voltage (+60mv) [0x6] voltage (+70mv) [0x7] voltage (+80mv) [0x8] voltage (+90mv) [0x9] voltage (+100mv) [0xA] voltage (-10mv) [0xB] voltage (-20mv) [0xC] voltage (-30mv) [0xD] voltage (-40mv) [0xE] voltage (-50mv) [0xF] voltage (-60mv) [0x10] voltage (-70mv) [0x11] voltage (-80mv) [0x12] voltage (-90mv) [0x13] voltage (-100mv) [0x14]
VDDQ voltage	Selection	voltage(0.5V) [0x0] ■ default voltage (+10mv) [0x1] voltage (+20mv) [0x2] voltage (+30mv) [0x3] voltage (+40mv) [0x4] voltage (+50mv) [0x5] voltage (+60mv) [0x6] voltage (+70mv) [0x7] voltage (+80mv) [0x8] voltage (+90mv) [0x9] voltage (+100mv) [0xA] voltage (-10mv) [0xB] voltage (-20mv) [0xC] voltage (-30mv) [0xD] voltage (-40mv) [0xE] voltage (-50mv) [0xF] voltage (-60mv) [0x10] voltage (-70mv) [0x11] voltage (-80mv) [0x12] voltage (-90mv) [0x13] voltage (-100mv) [0x14]

Setting	Type	Options / Values / Range
VDD MEM voltage	Selection	voltage(0.78V) [0x0] ■ default voltage (+10mv) [0x1] voltage (+20mv) [0x2] voltage (+30mv) [0x3] voltage (+40mv) [0x4] voltage (+50mv) [0x5] voltage (+60mv) [0x6] voltage (+70mv) [0x7] voltage (+80mv) [0x8] voltage (+90mv) [0x9] voltage (+100mv) [0xA] voltage (-10mv) [0xB] voltage (-20mv) [0xC] voltage (-30mv) [0xD] voltage (-40mv) [0xE] voltage (-50mv) [0xF] voltage (-60mv) [0x10] voltage (-70mv) [0x11] voltage (-80mv) [0x12] voltage (-90mv) [0x13] voltage (-100mv) [0x14]
POST LED on enable	Selection	Disabled [0x0] Enabled [0x1] ■ default
Smart Mux Support	Selection	Disabled [0x0] ■ default Hybrid Graphics Mode [0x1] Discrete Mode [0x2] Smart Mux 1.5 [0x3] Smart Mux 2.0 [0x4]
EC UDC Log Support	Selection	Disabled [0x0] Enabled [0x1] ■ default
Retiemr I2C0/SMBUS1 Enable	Selection	Disabled [0x0] ■ default Enabled [0x1]
Retimer FMP driver support	Selection	Disabled [0x0] ■ default I2c Retimer FMP driver without force power [0x1] I2c Retimer FMP driver with force power [0x2] SBU Retimer FMP driver without force power [0x3] SBU Retimer FMP driver with force power [0x4]

Debug Configurations (0x16)

Setting	Type	Options / Values / Range
Serial Debug Message Under OS	Selection	Disabled [0x0] ■ default Enabled [0x1]
Debug Print In ASL	Selection	Disabled [0x0] ■ default Enabled [0x1]

Setting	Type	Options / Values / Range
ISP Camera Module List	Selection	Disabled [0x0] ■ default Module 1 [0x1] Module 2 [0x2] Module 3 [0x3] Module 4 [0x4] Module 5 [0x5] Module 6 [0x6] Module 7 [0x7] Module 8 [0x8] Module 9 [0x9] Module 10 [0xA] Module 11 [0xB] Module 12 [0xC] Module 13 [0xD]
UEFI ISP I2c support	Selection	Disabled [0x0] ■ default Enabled [0x1]
Debug Data Preserve	Selection	Disabled [0x0] ■ default Enabled [0x1]
Time for Blank Screen (10ms)	Numeric	Min 0x0 · Max 0x3E8 · Step 0x1 · Default 0xC8 ■ default

AMD Overclocking (AOD)

28 forms · 153 settings

AMD Overclocking Option (0x98)

Submenus Accept ■ Decline

AMD Overclocking (0x100)

Submenus Manual CPU Overclocking ■ Manual iGPU Overclocking ■ DDR and Infinity Fabric Frequency/Timings ■ Precision Boost Overdrive ■ VDDP Voltage Control ■ SoC/Uncore OC Mode ■ SoC Voltage ■ LN2 Mode ■ VDD Misc ■ LCLK Frequency Control ■ Onboard Voltage Control

Manual CPU Overclocking (0x101)

Submenus CPU Core Count Control

Setting	Type	Options / Values / Range
CPU Frequency	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x5 · Default 0x0 ■ default ■ performance TUNE Read-only/Auto target. Use PBO + Curve Optimizer rather than a fixed all-core frequency (which disables opportunistic boost).
CPU Voltage	Numeric	Min 0x0 · Max 0xAF0 · Default 0x0 ■ default ■ performance TUNE Auto. Manual Vcore disables adaptive boost; undervolt via Curve Optimizer instead.

CPU Core Count Control (0x10C)

Setting	Type	Options / Values / Range
SMT Control	Selection	Auto [0x1] ■ default Disable [0x0] ■ performance KEEP Auto (SMT on). Simultaneous multithreading gives roughly 20-30% more multi-thread throughput; Disable only for niche per-core-latency cases.
Down Core Mode	Selection	Enable [0x1] Disable [0x0] ■ default ■ performance KEEP Disable. Down-coring reduces multi-thread performance; enable it only for special validation.

Manual iGPU Overclocking (0x102)

Setting	Type	Options / Values / Range
GFX Clock Frequency	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■ default ■ performance TUNE Auto. Fixing iGPU clock removes dynamic boost; prefer GPU boost-override + GFX curve.
GFX Voltage	Numeric	Min 0x0 · Max 0xAF0 · Default 0x0 ■ default ■ performance TUNE Auto. Manual iGPU voltage is expert-only; undervolt via GFX Curve Optimizer.

DDR and Infinity Fabric Frequency/Timings (0x103)

Submenus DDR Options Infinity Fabric Frequency and Dividers

DDR Options (0x110)Submenus [DDR Timing Configuration](#) [DDR Bus Configuration](#) [DDR Controller Configuration](#)**DDR Timing Configuration (0x112)**Submenus [DDR SPD Timing](#) [DDR Non-SPD Timing](#)

Setting	Type	Options / Values / Range
Active Memory Timing Settings	Selection	Auto [0xFF] ■ default Enabled [0x1]
Memory Target Speed	Selection	Auto [0xFFFF] ■ default 2000 MTs [0x7D0] 2400 MTs [0x960] 3200 MTs [0xC80] 3400 MTs [0xD48] 3600 MTs [0xE10] 3800 MTs [0xED8] 4000 MTs [0xFA0] 4200 MTs [0x1068] 4400 MTs [0x1130] 4600 MTs [0x11F8] 4800 MTs [0x12C0] 5000 MTs [0x1388] 5200 MTs [0x1450] 5400 MTs [0x1518] 5600 MTs [0x15E0] 5800 MTs [0x16A8] 6000 MTs [0x1770] 6200 MTs [0x1838] 6400 MTs [0x1900] 6600 MTs [0x19C8] 6800 MTs [0x1A90] 7000 MTs [0x1B58] 7200 MTs [0x1C20] 7400 MTs [0x1CE8] 7600 MTs [0x1DB0] 7800 MTs [0x1E78] 8000 MTs [0x1F40] 8200 MTs [0x2008] 8400 MTs [0x20D0] 8600 MTs [0x2198] 8800 MTs [0x2260] 9000 MTs [0x2328] 9200 MTs [0x23F0] 9400 MTs [0x24B8] 9600 MTs [0x2580] 9800 MTs [0x2648] 10000 MTs [0x2710] 10200 MTs [0x27D8] 10400 MTs [0x28A0] 10600 MTs [0x2968] 10800 MTs [0x2A30] 11000 MTs [0x2AF8] 11200 MTs [0x2BC0] 11400 MTs [0x2C88] 11600 MTs [0x2D50]

Setting	Type	Options / Values / Range
Memory Target Speed (cont.)		11800 MTs [0x2E18] 12000 MTs [0x2EE0] ■ performance TUNE Keep at the soldered LPDDR5X rated speed (8000 MT/s). This memory is not user-replaceable and already runs at spec.

DDR SPD Timing (0x11B)

Setting	Type	Options / Values / Range
Tcl Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Tcl	Numeric	Min 0x16 · Max 0x3E · Step 0x2 · Default 0x16 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Trcd Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Trcd	Numeric	Min 0x8 · Max 0x3E · Default 0x8 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Trp Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Trp	Numeric	Min 0x8 · Max 0x3E · Default 0x8 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Tras Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.

Setting	Type	Options / Values / Range
Tras	Numeric	Min 0x1E · Max 0x7E · Default 0x1E ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Trc Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Trc	Numeric	Min 0x20 · Max 0xFF · Default 0x20 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Twr Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Twr	Numeric	Min 0x30 · Max 0x60 · Default 0x30 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Trfc1 Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Trfc1	Numeric	Min 0x32 · Max 0xFFF · Default 0x32 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Trfc2 Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Trfc2	Numeric	Min 0x32 · Max 0xFFF · Default 0x32 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.

Setting	Type	Options / Values / Range
TrfcSb Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TrfcSb	Numeric	Min 0x32 · Max 0x7FF · Default 0x32 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Trtp Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Trtp	Numeric	Min 0x5 · Max 0x1F · Default 0x9 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TrrdL Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TrrdL	Numeric	Min 0x4 · Max 0x20 · Default 0x4 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TrrdS Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TrrdS	Numeric	Min 0x4 · Max 0x14 · Default 0x4 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Tfaw Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.

Setting	Type	Options / Values / Range
Tfaw	Numeric	Min 0x14 · Max 0x50 · Default 0x1A ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TwtrL Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TwtrL	Numeric	Min 0x8 · Max 0x30 · Default 0x8 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TwtrS Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TwtrS	Numeric	Min 0x2 · Max 0x10 · Default 0x3 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.

DDR Non-SPD Timing (0x11C)

Setting	Type	Options / Values / Range
TrdrdScL Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TrdrdScL	Numeric	Min 0x1 · Max 0xF · Default 0x1 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TrdrdSc Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.

Setting	Type	Options / Values / Range
TrdrdSc	Numeric	Min 0x1 · Max 0xF · Default 0x1 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TrdrdSd Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TrdrdSd	Numeric	Min 0x1 · Max 0xF · Default 0x3 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TrdrdDd Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TrdrdDd	Numeric	Min 0x1 · Max 0xF · Default 0x3 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TwrrwrScL Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TwrrwrScL	Numeric	Min 0x1 · Max 0x3F · Default 0x1 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TwrrwrSc Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TwrrwrSc	Numeric	Min 0x1 · Max 0xF · Default 0x1 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.

Setting	Type	Options / Values / Range
TwrrrSd Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TwrrrSd	Numeric	Min 0x1 · Max 0xF · Default 0x3 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TwrrrDd Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
TwrrrDd	Numeric	Min 0x1 · Max 0xF · Default 0x3 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Twrrd Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Twrrd	Numeric	Min 0x1 · Max 0xF · Default 0x5 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Trdwr Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.
Trdwr	Numeric	Min 0x1 · Max 0x1F · Default 0x5 ■ default ■ performance TUNE DRAM sub-timing. On soldered LPDDR5X these are factory-trained to spec; tightening risks instability with little headroom. Leave Auto unless you are an experienced memory overclocker validating each change.

DDR Bus Configuration (0x113)

Setting	Type	Options / Values / Range
Processor CA drive strengths	Selection	Auto [0xFF] ■ default 120.0 Ohm [0x78] 60.0 Ohm [0x3C] 40.0 Ohm [0x28] 30.0 Ohm [0x1E]
Processor DQ drive strengths	Selection	Auto [0xFF] ■ default High Impedance [0x0] 240 ohm [0x2] 120 ohm [0x4] 80 ohm [0x6] 60 ohm [0xC] 48 ohm [0xE] 40 ohm [0x1C] 34.3 ohm [0x1E]
Processor ODT impedance	Selection	Auto [0xFF] ■ default High Impedance [0x0] 480 ohm [0x1] 240 ohm [0x2] 160 ohm [0x3] 120 ohm [0x4] 96 ohm [0x5] 80 ohm [0x6] 68.6 ohm [0x7] 60 ohm [0xC] 53.3 ohm [0xD] 48 ohm [0xE] 43.6 ohm [0xF] 40 ohm [0x1C] 36.9 ohm [0x1D] 34.3 ohm [0x1E] 32 ohm [0x1F] 30 ohm [0x3C] 28.2 ohm [0x3D] 26.7 ohm [0x3E] 25.3 ohm [0x3F]
Dram DQ drive strengths	Selection	Auto [0xFF] ■ default 48 ohm [0x2] 40 ohm [0x1] 34 ohm [0x0]
Dram ODT impedance RTT_NOM_WR	Selection	Auto [0xFF] ■ default RTT_OFF [0x0] RZQ (240) [0x1] RZQ/2 (120) [0x2] RZQ/3 (80) [0x3] RZQ/4 (60) [0x4] RZQ/5 (48) [0x5] RZQ/6 (40) [0x6] RZQ/7 (34) [0x7]

Setting	Type	Options / Values / Range
Dram ODT impedance RTT_NOM_RD	Selection	Auto [0xFF] ■ default RTT_OFF [0x0] RZQ (240) [0x1] RZQ/2 (120) [0x2] RZQ/3 (80) [0x3] RZQ/4 (60) [0x4] RZQ/5 (48) [0x5] RZQ/6 (40) [0x6] RZQ/7 (34) [0x7]
Dram ODT impedance RTT_WR	Selection	Auto [0xFF] ■ default RTT_OFF [0x0] RZQ (240) [0x1] RZQ/2 (120) [0x2] RZQ/3 (80) [0x3] RZQ/4 (60) [0x4] RZQ/5 (48) [0x5] RZQ/6 (40) [0x6] RZQ/7 (34) [0x7]
Dram ODT impedance RTT_PARK	Selection	Auto [0xFF] ■ default RTT_OFF [0x0] RZQ (240) [0x1] RZQ/2 (120) [0x2] RZQ/3 (80) [0x3] RZQ/4 (60) [0x4] RZQ/5 (48) [0x5] RZQ/6 (40) [0x6] RZQ/7 (34) [0x7]
Dram ODT impedance DQS_RTT_PARK	Selection	Auto [0xFF] ■ default RTT_OFF [0x0] RZQ (240) [0x1] RZQ/2 (120) [0x2] RZQ/3 (80) [0x3] RZQ/4 (60) [0x4] RZQ/5 (48) [0x5] RZQ/6 (40) [0x6] RZQ/7 (34) [0x7]

DDR Controller Configuration (0x114)

Submenus DDR Power Options

DDR Power Options (0x11D)

Setting	Type	Options / Values / Range
Power Down Enable	Selection	Disabled [0x0] Enabled [0x1] Auto [0xFF] ■ default ■ performance TUNE Disabled can trim memory latency slightly under load; Enabled (Auto) saves idle power. Default is fine for most.

Infinity Fabric Frequency and Dividers (0x111)

Setting	Type	Options / Values / Range
Infinity Fabric Frequency and Dividers	Selection	Auto [0xFFFF] ■ default 333 MHz [0x14D] 400 MHz [0x190] 533 MHz [0x215] 667 MHz [0x29B] 800 MHz [0x320] 933 MHz [0x3A5] 1050 MHz [0x41A] 1066 MHz [0x42A] 1067 MHz [0x42B] 1200 MHz [0x4B0] 1333 MHz [0x535] 1367 MHz [0x557] 1400 MHz [0x578] 1433 MHz [0x599] 1467 MHz [0x5BB] 1500 MHz [0x5DC] 1533 MHz [0x5FD] 1567 MHz [0x61F] 1600 MHz [0x640] 1633 MHz [0x661] 1667 MHz [0x683] 1700 MHz [0x6A4] 1733 MHz [0x6C5] 1767 MHz [0x6E7] 1800 MHz [0x708] 1833 MHz [0x729] 1867 MHz [0x74B] 1900 MHz [0x76C] 1933 MHz [0x78D] 1967 MHz [0x7AF] 2000 MHz [0x7D0] 2033 MHz [0x7F1] 2067 MHz [0x813] 2100 MHz [0x834] 2133 MHz [0x855] 2167 MHz [0x877] 2200 MHz [0x898] 2233 MHz [0x8B9] 2267 MHz [0x8DB] 2300 MHz [0x8FC] 2333 MHz [0x91D] 2367 MHz [0x93F] 2400 MHz [0x960] 2433 MHz [0x981] 2467 MHz [0x9A3]

Setting	Type	Options / Values / Range
Infinity Fabric Frequency and Dividers (<i>cont.</i>)		2500 MHz [0x9C4] 2550 MHz [0x9F6] 2600 MHz [0xA28] 2650 MHz [0xA5A] 2700 MHz [0xA8C] 2750 MHz [0xABE] 2800 MHz [0xAF0] 2850 MHz [0xB22] 2900 MHz [0xB54] 2950 MHz [0xB86] 3000 MHz [0xBB8] ■ performance TUNE Auto. On Strix Halo the fabric/UCLK ratio is pre-balanced to the soldered LPDDR5X; manual changes rarely help and can destabilize.
UCLK DIV1 MODE	Selection	Auto [0xFF] ■ default UCLK=MEMCLK [0x1] UCLK=MEMCLK/2 [0x0] ■ performance TUNE Auto (UCLK=MEMCLK, i.e. 1:1, where firmware allows it). 1:1 minimizes memory latency; leave the firmware default for soldered RAM.

Precision Boost Overdrive (0x104)

Submenus GFX Curve Optimizer Curve Optimizer Curve Shaper

Setting	Type	Options / Values / Range
Precision Boost Overdrive	Selection	Auto [0xFF] ■ default Disable [0x0] Enabled [0x1] Advanced [0x2] ■ performance CHANGE Advanced (or Enabled). Raises PPT/TDC/EDC ceilings so boost sustains longer — the main free-performance lever. Pair with adequate cooling.
PBO Limits	Selection	Auto [0xFF] ■ default Disable [0x0] Motherboard [0x1] Manual [0x2] ■ performance CHANGE Motherboard (not Auto) to let the chip use the full board power budget, or Manual to set explicit PPT/TDC/EDC. Auto is conservative.
PPT Limit [mW]	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ default ■ performance TUNE Raise toward the platform max (this APU sustains up to ~140 W on the GTR9 Pro) for more all-core boost, within thermal limits.
TDC Limit [mA]	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ default ■ performance TUNE Increase alongside PPT for sustained-load current headroom; keep within VRM/thermal capability.
EDC Limit [mA]	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ default ■ performance TUNE Increase for peak/burst current headroom; raising EDC mainly helps short bursty boost.
CCX (or second rail) TDC Limit [mA]	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ default ■ performance TUNE Secondary-rail TDC — raise in step with the main TDC limit when pushing all-core.

Setting	Type	Options / Values / Range
CCX (or second rail) EDC Limit [mA]	Numeric	Min 0x0 · Max 0xFFFFFFFF · Default 0x0 ■ default ■ performance TUNE Secondary-rail EDC — raise in step with the main EDC limit for burst headroom.
Precision Boost Overdrive Scalar Ctrl	Selection	Auto [0xFF] ■ default Manual [0x1] ■ performance TUNE Manual to expose the Scalar. Higher scalar extends boost residency (more heat); start conservative.
Precision Boost Overdrive Scalar	Selection	1X [0x64] ■ default 2X [0xC8] 3X [0x12C] 4X [0x190] 5X [0x1F4] 6X [0x258] 7X [0x2BC] 8X [0x320] 9X [0x384] 10X [0x3E8] ■ performance TUNE 1X-2X for daily; up to 10X sustains higher voltage/boost longer at the cost of heat and longevity. 2-4X is a sane gaming target.
CPU Boost Clock Override	Selection	Disabled [0x0] ■ default Enabled (Positive) [0x1] Enabled (Negative) [0x2] ■ performance TUNE Enabled (Positive) to allow up to +200 MHz above rated boost when thermals permit.
Max CPU Boost Clock Override(+)	Numeric	Min 0x19 · Max 0xC8 · Step 0x19 · Default 0x19 ■ default ■ performance TUNE Add up to +200 MHz; effective only with thermal/power headroom (PBO + cooling).
Max CPU Boost Clock Override(-)	Numeric	Min 0x19 · Max 0x3E8 · Step 0x19 · Default 0x19 ■ default
GPU Boost Clock Override	Selection	Enabled [0x1] Disabled [0x0] ■ default ■ performance TUNE Enabled (Positive) to lift the iGPU boost ceiling for graphics-bound gains, thermals permitting.
Max GPU Boost Clock Override	Selection	0 MHz [0x0] ■ default 25 MHz [0x19] 50 MHz [0x32] 75 MHz [0x4B] 100 MHz [0x64] 125 MHz [0x7D] 150 MHz [0x96] 175 MHz [0xAF] 200 MHz [0xC8] ■ performance TUNE Add a modest iGPU boost offset and test; the 8060S is bandwidth/thermal-limited, so gains taper.
Platform Thermal Throttle Ctrl	Selection	Manual [0x1] Auto [0x0] ■ default ■ performance TUNE Manual to raise the throttle limit for more sustained clocks (only with strong cooling).

Setting	Type	Options / Values / Range
Platform Thermal Throttle Limit	Numeric	Min 0x0 · Max 0xFF · Default 0x0 ■ default ■ performance TUNE Raising the skin/platform throttle point sustains boost longer but increases temperatures and fan noise.

GFX Curve Optimizer (0x11E)

Setting	Type	Options / Values / Range
GFX Curve Optimizer	Selection	Disable [0x0] ■ default GFX Curve Optimizer [0x1] ■ performance TUNE GFX Curve Optimizer enables the iGPU offset; pair it with a Negative sign below to undervolt for more sustained graphics clock within the shared power budget.
GFX Curve Optimizer Sign	Selection	Positive [0x0] ■ default Negative [0x1] ■ performance TUNE Negative undervolts the iGPU V/F curve for cooler, higher-sustained clocks.
GFX Curve Optimizer Magnitude	Numeric	Min 0x0 · Max 0x32 · Default 0x0 ■ default ■ performance TUNE Start small (e.g. -10) and validate; iGPU stability is workload-sensitive.

Curve Optimizer (0x11F)

Setting	Type	Options / Values / Range
Curve Optimizer	Selection	Disable [0x0] ■ default All Cores [0x1] Per Core [0x2] ■ performance CHANGE All Cores, with a Negative sign below. Undervolting the V/F curve lets the chip hit higher effective clocks cooler - the highest-value PBO sub-setting.
All Core Curve Optimizer Sign	Selection	Positive [0x0] ■ default Negative [0x1] ■ performance CHANGE Negative. A negative offset undervolts for higher sustained boost and lower temps.
All Core Curve Optimizer Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ default ■ performance TUNE Start around -15 to -20 and stress-test; stable mobile Zen 5 silicon often reaches -25 to -30. Validate for stability.

Curve Shaper (0x126)

Setting	Type	Options / Values / Range
Min Frequency — Low Temperature	Selection	Auto [0xFF] ■ default Enable [0x1] Disable [0x0] ■ performance TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region. Use it to refine a Curve Optimizer tune, not to replace one.

Setting	Type	Options / Values / Range
Min Frequency — Low Temperature Sign	Selection	Positive [0x0] ■ default Negative [0x1] ■ performance TUNE Negative undervolts this frequency/temperature band. Hot and high-frequency bands generally tolerate a smaller negative offset than cold, low-frequency ones.
Min Frequency — Low Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ default ■ performance TUNE Offset magnitude for this band (max 30). Start at 3-5 per band and validate; Curve Shaper sharpens an existing Curve Optimizer result rather than substituting for it.
Min Frequency — Med Temperature	Selection	Auto [0xFF] ■ default Enable [0x1] Disable [0x0] ■ performance TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region. Use it to refine a Curve Optimizer tune, not to replace one.
Min Frequency — Med Temperature Sign	Selection	Positive [0x0] ■ default Negative [0x1] ■ performance TUNE Negative undervolts this frequency/temperature band. Hot and high-frequency bands generally tolerate a smaller negative offset than cold, low-frequency ones.
Min Frequency — Med Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ default ■ performance TUNE Offset magnitude for this band (max 30). Start at 3-5 per band and validate; Curve Shaper sharpens an existing Curve Optimizer result rather than substituting for it.
Min Frequency — High Temperature	Selection	Auto [0xFF] ■ default Enable [0x1] Disable [0x0] ■ performance TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region. Use it to refine a Curve Optimizer tune, not to replace one.
Min Frequency — High Temperature Sign	Selection	Positive [0x0] ■ default Negative [0x1] ■ performance TUNE Negative undervolts this frequency/temperature band. Hot and high-frequency bands generally tolerate a smaller negative offset than cold, low-frequency ones.
Min Frequency — High Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ default ■ performance TUNE Offset magnitude for this band (max 30). Start at 3-5 per band and validate; Curve Shaper sharpens an existing Curve Optimizer result rather than substituting for it.
Low Frequency — Low Temperature	Selection	Auto [0xFF] ■ default Enable [0x1] Disable [0x0] ■ performance TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region. Use it to refine a Curve Optimizer tune, not to replace one.
Low Frequency — Low Temperature Sign	Selection	Positive [0x0] ■ default Negative [0x1] ■ performance TUNE Negative undervolts this frequency/temperature band. Hot and high-frequency bands generally tolerate a smaller negative offset than cold, low-frequency ones.

Setting	Type	Options / Values / Range
Low Frequency — Low Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ default ■ performance TUNE Offset magnitude for this band (max 30). Start at 3-5 per band and validate; Curve Shaper sharpens an existing Curve Optimizer result rather than substituting for it.
Low Frequency — Med Temperature	Selection	Auto [0xFF] ■ default Enable [0x1] Disable [0x0] ■ performance TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region. Use it to refine a Curve Optimizer tune, not to replace one.
Low Frequency — Med Temperature Sign	Selection	Positive [0x0] ■ default Negative [0x1] ■ performance TUNE Negative undervolts this frequency/temperature band. Hot and high-frequency bands generally tolerate a smaller negative offset than cold, low-frequency ones.
Low Frequency — Med Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ default ■ performance TUNE Offset magnitude for this band (max 30). Start at 3-5 per band and validate; Curve Shaper sharpens an existing Curve Optimizer result rather than substituting for it.
Low Frequency — High Temperature	Selection	Auto [0xFF] ■ default Enable [0x1] Disable [0x0] ■ performance TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region. Use it to refine a Curve Optimizer tune, not to replace one.
Low Frequency — High Temperature Sign	Selection	Positive [0x0] ■ default Negative [0x1] ■ performance TUNE Negative undervolts this frequency/temperature band. Hot and high-frequency bands generally tolerate a smaller negative offset than cold, low-frequency ones.
Low Frequency — High Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ default ■ performance TUNE Offset magnitude for this band (max 30). Start at 3-5 per band and validate; Curve Shaper sharpens an existing Curve Optimizer result rather than substituting for it.
Med Frequency — Low Temperature	Selection	Auto [0xFF] ■ default Enable [0x1] Disable [0x0] ■ performance TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region. Use it to refine a Curve Optimizer tune, not to replace one.
Med Frequency — Low Temperature Sign	Selection	Positive [0x0] ■ default Negative [0x1] ■ performance TUNE Negative undervolts this frequency/temperature band. Hot and high-frequency bands generally tolerate a smaller negative offset than cold, low-frequency ones.
Med Frequency — Low Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ default ■ performance TUNE Offset magnitude for this band (max 30). Start at 3-5 per band and validate; Curve Shaper sharpens an existing Curve Optimizer result rather than substituting for it.

Setting	Type	Options / Values / Range
Med Frequency — Med Temperature	Selection	Auto [0xFF] ■ default Enable [0x1] Disable [0x0] ■ performance TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region. Use it to refine a Curve Optimizer tune, not to replace one.
Med Frequency — Med Temperature Sign	Selection	Positive [0x0] ■ default Negative [0x1] ■ performance TUNE Negative undervolts this frequency/temperature band. Hot and high-frequency bands generally tolerate a smaller negative offset than cold, low-frequency ones.
Med Frequency — Med Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ default ■ performance TUNE Offset magnitude for this band (max 30). Start at 3-5 per band and validate; Curve Shaper sharpens an existing Curve Optimizer result rather than substituting for it.
Med Frequency — High Temperature	Selection	Auto [0xFF] ■ default Enable [0x1] Disable [0x0] ■ performance TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region. Use it to refine a Curve Optimizer tune, not to replace one.
Med Frequency — High Temperature Sign	Selection	Positive [0x0] ■ default Negative [0x1] ■ performance TUNE Negative undervolts this frequency/temperature band. Hot and high-frequency bands generally tolerate a smaller negative offset than cold, low-frequency ones.
Med Frequency — High Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ default ■ performance TUNE Offset magnitude for this band (max 30). Start at 3-5 per band and validate; Curve Shaper sharpens an existing Curve Optimizer result rather than substituting for it.
High Frequency — Low Temperature	Selection	Auto [0xFF] ■ default Enable [0x1] Disable [0x0] ■ performance TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region. Use it to refine a Curve Optimizer tune, not to replace one.
High Frequency — Low Temperature Sign	Selection	Positive [0x0] ■ default Negative [0x1] ■ performance TUNE Negative undervolts this frequency/temperature band. Hot and high-frequency bands generally tolerate a smaller negative offset than cold, low-frequency ones.
High Frequency — Low Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ default ■ performance TUNE Offset magnitude for this band (max 30). Start at 3-5 per band and validate; Curve Shaper sharpens an existing Curve Optimizer result rather than substituting for it.
High Frequency — Med Temperature	Selection	Auto [0xFF] ■ default Enable [0x1] Disable [0x0] ■ performance TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region. Use it to refine a Curve Optimizer tune, not to replace one.

Setting	Type	Options / Values / Range
High Frequency — Med Temperature Sign	Selection	Positive [0x0] ■ default Negative [0x1] ■ performance TUNE Negative undervolts this frequency/temperature band. Hot and high-frequency bands generally tolerate a smaller negative offset than cold, low-frequency ones.
High Frequency — Med Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ default ■ performance TUNE Offset magnitude for this band (max 30). Start at 3-5 per band and validate; Curve Shaper sharpens an existing Curve Optimizer result rather than substituting for it.
High Frequency — High Temperature	Selection	Auto [0xFF] ■ default Enable [0x1] Disable [0x0] ■ performance TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region. Use it to refine a Curve Optimizer tune, not to replace one.
High Frequency — High Temperature Sign	Selection	Positive [0x0] ■ default Negative [0x1] ■ performance TUNE Negative undervolts this frequency/temperature band. Hot and high-frequency bands generally tolerate a smaller negative offset than cold, low-frequency ones.
High Frequency — High Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ default ■ performance TUNE Offset magnitude for this band (max 30). Start at 3-5 per band and validate; Curve Shaper sharpens an existing Curve Optimizer result rather than substituting for it.
Max Frequency — Low Temperature	Selection	Auto [0xFF] ■ default Enable [0x1] Disable [0x0] ■ performance TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region. Use it to refine a Curve Optimizer tune, not to replace one.
Max Frequency — Low Temperature Sign	Selection	Positive [0x0] ■ default Negative [0x1] ■ performance TUNE Negative undervolts this frequency/temperature band. Hot and high-frequency bands generally tolerate a smaller negative offset than cold, low-frequency ones.
Max Frequency — Low Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ default ■ performance TUNE Offset magnitude for this band (max 30). Start at 3-5 per band and validate; Curve Shaper sharpens an existing Curve Optimizer result rather than substituting for it.
Max Frequency — Med Temperature	Selection	Auto [0xFF] ■ default Enable [0x1] Disable [0x0] ■ performance TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region. Use it to refine a Curve Optimizer tune, not to replace one.
Max Frequency — Med Temperature Sign	Selection	Positive [0x0] ■ default Negative [0x1] ■ performance TUNE Negative undervolts this frequency/temperature band. Hot and high-frequency bands generally tolerate a smaller negative offset than cold, low-frequency ones.

Setting	Type	Options / Values / Range
Max Frequency — Med Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ default ■ performance TUNE Offset magnitude for this band (max 30). Start at 3-5 per band and validate; Curve Shaper sharpens an existing Curve Optimizer result rather than substituting for it.
Max Frequency — High Temperature	Selection	Auto [0xFF] ■ default Enable [0x1] Disable [0x0] ■ performance TUNE Curve Shaper point (Zen 5 PBO2). Enables a per-band V/F offset for this frequency/temperature region. Use it to refine a Curve Optimizer tune, not to replace one.
Max Frequency — High Temperature Sign	Selection	Positive [0x0] ■ default Negative [0x1] ■ performance TUNE Negative undervolts this frequency/temperature band. Hot and high-frequency bands generally tolerate a smaller negative offset than cold, low-frequency ones.
Max Frequency — High Temperature Magnitude	Numeric	Min 0x0 · Max 0x1E · Default 0x0 ■ default ■ performance TUNE Offset magnitude for this band (max 30). Start at 3-5 per band and validate; Curve Shaper sharpens an existing Curve Optimizer result rather than substituting for it.

VDDP Voltage Control (0x105)

Setting	Type	Options / Values / Range
VDDP Voltage Control	Selection	Auto [0xFF] ■ default Manual [0x1]
VDDP Voltage Adjust	Numeric	Min 0x0 · Max 0x7D0 · Default 0x0 ■ default

SoC/Uncore OC Mode (0x106)

Setting	Type	Options / Values / Range
SoC/Uncore OC Mode	Selection	Auto [0xFF] ■ default Enabled [0x1] Disabled [0x0] ■ performance TUNE Auto. Forcing max SoC/uncore raises power and heat for little gain on this APU; leave default.

SoC Voltage (0x107)

Setting	Type	Options / Values / Range
SoC Voltage	Numeric	Min 0x0 · Max 0xAF0 · Default 0x0 ■ default

LN2 Mode (0x108)

Setting	Type	Options / Values / Range
LN2 Mode	Selection	Auto [0xFF] ■ default Disabled [0x0] Enabled [0x1]

VDD Misc (0x109)

Setting	Type	Options / Values / Range
VDD Misc Control	Selection	Auto [0x0] ■ default Manual [0x1]
VDD Misc Voltage	Numeric	Min 0x1F4 · Max 0x15E0 · Default 0x320 ■ default

LCLK Frequency Control (0x10A)

Setting	Type	Options / Values / Range
LCLK Frequency Control	Selection	Auto [0x0] ■ default Manual [0x1] ■ performance TUNE Auto. LCLK (NBIO/PCIe fabric clock) — Manual max can lower PCIe/IO latency but raises power; default is balanced.
Minimum Frequency	Numeric	Min 0x96 · Max 0x9C4 · Default 0x96 ■ default ■ performance TUNE LCLK floor. Raising the minimum keeps the NBIO/PCIe fabric clock high for lower IO latency at the cost of idle power; default is balanced.
Maximum Frequency	Numeric	Min 0x96 · Max 0x9C4 · Default 0x9C4 ■ default ■ performance TUNE LCLK ceiling. Leaving at max lets the IO fabric reach full speed under load; default Auto already allows this.

Onboard Voltage Control (0x10B)

Submenus VDDIO Voltage Control ■ VPP Voltage Control ■ VDD_MEM Voltage Control

VDDIO Voltage Control (0x123)

Information fields Current DIMM VDD Voltage, Current DIMM VDDQ Voltage, Current APU VDDIO Voltage

Setting	Type	Options / Values / Range
VDDIO Ctrl	Selection	Auto [0x0] ■ default Manual [0x1] Separate [0x2]
DIMM VDD Adjust	Numeric	Min 0x320 · Max 0x596 · Step 0xA · Default 0x44C ■ default
DIMM VDDQ Adjust	Numeric	Min 0x320 · Max 0x596 · Default 0x44C ■ default

VPP Voltage Control (0x124)

Information fields Current VPP Voltage

Setting	Type	Options / Values / Range
VPP Ctrl	Selection	Auto [0x0] ■ default Manual [0x1]
VPP Adjust	Numeric	Min 0x5DC · Max 0x852 · Step 0xA · Default 0x708 ■ default

VDD_MEM Voltage Control (0x125)

Information fields Current VDD_MEM Voltage

Setting	Type	Options / Values / Range
VDD_MEM Control	Selection	Auto [0x0] ■ default Manual [0x1]

AMD PMF (Platform Management Framework)

28 forms · 139 settings

AMD PMF Feature (0x64)

Submenus APMF Multiple Binaries Settings APTS Control Method Settings ATST Function APMF Function 1 Settings ■■■■ APMF Function 4 Settings APMF Function 7 Settings APMF Function 9 Settings APMF Function 11 Settings APMF ■■ Function 12 Settings APMF Function 14 Settings APMF Function 16 Settings

Setting	Type	Options / Values / Range
PMF Device Support	Selection	Disabled [0x0] ■ default Enabled [0x1]

APMF Multiple Binaries Settings (0x6D)

Setting	Type	Options / Values / Range
APMF Multiple Binaries Support	Selection	Disabled [0x0] ■ default Enabled [0x1]
PMF Policy Binaries Index	Selection	PMF Binary 0 [0x0] ■ default PMF Binary 1 [0x1] PMF Binary 2 [0x2] PMF Binary 3 [0x3] PMF Binary 4 [0x4] PMF Binary 5 [0x5] PMF Binary 6 [0x6] PMF Binary 7 [0x7] PMF Binary 8 [0x8] PMF Binary 9 [0x9]
PMF Binary 0 Size	Selection	0 KB [0x0] ■ default 30 KB [0x1] 60 KB [0x2] 90 KB [0x3] 120 KB [0x4] 150 KB [0x5] 180 KB [0x6] 210 KB [0x7] 240 KB [0x8] 270 KB [0x9] 300 KB [0xA]
PMF Binary 1 Size	Selection	0 KB [0x0] ■ default 30 KB [0x1] 60 KB [0x2] 90 KB [0x3] 120 KB [0x4] 150 KB [0x5] 180 KB [0x6] 210 KB [0x7] 240 KB [0x8] 270 KB [0x9] 300 KB [0xA]

Setting	Type	Options / Values / Range
PMF Binary 2 Size	Selection	0 KB [0x0] ■ default 30 KB [0x1] 60 KB [0x2] 90 KB [0x3] 120 KB [0x4] 150 KB [0x5] 180 KB [0x6] 210 KB [0x7] 240 KB [0x8] 270 KB [0x9] 300 KB [0xA]
PMF Binary 3 Size	Selection	0 KB [0x0] ■ default 30 KB [0x1] 60 KB [0x2] 90 KB [0x3] 120 KB [0x4] 150 KB [0x5] 180 KB [0x6] 210 KB [0x7] 240 KB [0x8] 270 KB [0x9] 300 KB [0xA]
PMF Binary 4 Size	Selection	0 KB [0x0] ■ default 30 KB [0x1] 60 KB [0x2] 90 KB [0x3] 120 KB [0x4] 150 KB [0x5] 180 KB [0x6] 210 KB [0x7] 240 KB [0x8] 270 KB [0x9] 300 KB [0xA]
PMF Binary 5 Size	Selection	0 KB [0x0] ■ default 30 KB [0x1] 60 KB [0x2] 90 KB [0x3] 120 KB [0x4] 150 KB [0x5] 180 KB [0x6] 210 KB [0x7] 240 KB [0x8] 270 KB [0x9] 300 KB [0xA]
PMF Binary 6 Size	Selection	0 KB [0x0] ■ default 30 KB [0x1] 60 KB [0x2] 90 KB [0x3] 120 KB [0x4] 150 KB [0x5] 180 KB [0x6] 210 KB [0x7] 240 KB [0x8] 270 KB [0x9] 300 KB [0xA]

Setting	Type	Options / Values / Range
PMF Binary 7 Size	Selection	0 KB [0x0] ■ default 30 KB [0x1] 60 KB [0x2] 90 KB [0x3] 120 KB [0x4] 150 KB [0x5] 180 KB [0x6] 210 KB [0x7] 240 KB [0x8] 270 KB [0x9] 300 KB [0xA]
PMF Binary 8 Size	Selection	0 KB [0x0] ■ default 30 KB [0x1] 60 KB [0x2] 90 KB [0x3] 120 KB [0x4] 150 KB [0x5] 180 KB [0x6] 210 KB [0x7] 240 KB [0x8] 270 KB [0x9] 300 KB [0xA]
PMF Binary 9 Size	Selection	0 KB [0x0] ■ default 30 KB [0x1] 60 KB [0x2] 90 KB [0x3] 120 KB [0x4] 150 KB [0x5] 180 KB [0x6] 210 KB [0x7] 240 KB [0x8] 270 KB [0x9] 300 KB [0xA]

APTS Control Method Settings (0x6E)

Submenus APTS State Index 0 ■ APTS State Index 1 ■ APTS State Index 2 ■ APTS State Index 3 ■ APTS State Index 4 ■ APTS State Index 5 ■ APTS State Index 6 ■ APTS State Index 7 ■ APTS State Index 8 ■ APTS State Index 9 ■ APTS State Index 10 ■ APTS State Index 11 ■ APTS State Index 12 ■ APTS State Index 13 ■ APTS State Index 14 ■ APTS State Index 15

Setting	Type	Options / Values / Range
Performance and Thermal State Control Method	Selection	Disabled [0x0] ■ default Enabled [0x1]

ATST Function (0x7F)

Setting	Type	Options / Values / Range
ATST Function 3	Selection	Disabled [0x0] ■ default Enabled [0x1]

APMF Function 1 Settings (0x65)

Setting	Type	Options / Values / Range
Heartbeat Time Interval	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x3C ■ default

Setting	Type	Options / Values / Range
PMF PPT Time Constant Control	Selection	Disabled [0x0] ■ default Enabled [0x1]
PMF PPT Time Constant	Numeric	Min 0x64 · Max 0x7530 · Step 0x1 · Default 0x64 ■ default

APMF Function 4 Settings (0x66)

Setting	Type	Options / Values / Range
AMPF Function 4	Selection	Disabled [0x0] ■ default Enabled [0x1]

APMF Function 7 Settings (0x67)

Setting	Type	Options / Values / Range
AMPF Function 7	Selection	Disabled [0x0] ■ default Enabled [0x1]

APMF Function 9 Settings (0x68)

Setting	Type	Options / Values / Range
AMPF Function 9	Selection	Disabled [0x0] ■ default Enabled [0x1]
APTS State Index of AC Best Performance	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x0 ■ default
APTS State Index of AC Best Balanced	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x1 ■ default
APTS State Index of AC Best Power Efficiency	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x2 ■ default
APTS State Index of AC Energy Saver	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x3 ■ default
APTS State Index of DC Best Performance	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x4 ■ default
APTS State Index of DC Best Balanced	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x5 ■ default
APTS State Index of DC Best Power Efficiency	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x6 ■ default
APTS State Index of DC Battery Saver	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x7 ■ default ■ performance TUNE Selects which APTS state applies on DC. Irrelevant on this mains-powered mini-PC - the AC state index is the one in effect.

APMF Function 11 Settings (0x69)

Setting	Type	Options / Values / Range
AMPF Function 11	Selection	Disabled [0x0] ■ default Enabled [0x1]
Flags	Numeric	Min 0x0 · Max 0xFFFF · Step 0x1 · Default 0x0 ■ default

Setting	Type	Options / Values / Range
APTS State Index of ASM Performance	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x0 ■ default
APTS State Index of ASM Balanced	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x1 ■ default
APTS State Index of ASM Virtually Silent	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x2 ■ default
APTS State Index of ASM Non-Interactive	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x3 ■ default

APMF Function 12 Settings (0x6A)

Setting	Type	Options / Values / Range
APMF Function 12	Selection	Disabled [0x0] ■ default Enabled [0x1]
Flags	Numeric	Min 0x0 · Max 0xFFFF · Step 0x1 · Default 0x0 ■ default
APTS State Index of ASM Performance	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x4 ■ default
APTS State Index of ASM Balanced	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x5 ■ default
APTS State Index of ASM Virtually Silent	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x6 ■ default
APTS State Index of ASM Non-Interactive	Numeric	Min 0x0 · Max 0xF · Step 0x1 · Default 0x7 ■ default

APMF Function 14 Settings (0x6B)

Setting	Type	Options / Values / Range
APMF Function 14	Selection	Disabled [0x0] ■ default Enabled [0x1]

APMF Function 16 Settings (0x6C)

Setting	Type	Options / Values / Range
APMF Function 16	Selection	Disabled [0x0] ■ default Enabled [0x1]

APTS State Index 0 (0x6F)

Setting	Type	Options / Values / Range
Fan Table Index	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE Selects which fan table this APTS state uses. Point it at a more aggressive table for sustained load; it does not itself define curve points.
PPT Limit for PMF power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default
PPT Limit for PMF APU Only power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default

Setting	Type	Options / Values / Range
STT Min Power Limit in milliwatts	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit APU in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit HS2 in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.

APTS State Index 1 (0x70)

Setting	Type	Options / Values / Range
Fan Table Index	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE Selects which fan table this APTS state uses. Point it at a more aggressive table for sustained load; it does not itself define curve points.
PPT Limit for PMF power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default
PPT Limit for PMF APU Only power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default
STT Min Power Limit in milliwatts	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit APU in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit HS2 in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.

APTS State Index 2 (0x71)

Setting	Type	Options / Values / Range
Fan Table Index	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE Selects which fan table this APTS state uses. Point it at a more aggressive table for sustained load; it does not itself define curve points.
PPT Limit for PMF power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default
PPT Limit for PMF APU Only power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default

Setting	Type	Options / Values / Range
STT Min Power Limit in milliwatts	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit APU in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit HS2 in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.

APTS State Index 3 (0x72)

Setting	Type	Options / Values / Range
Fan Table Index	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE Selects which fan table this APTS state uses. Point it at a more aggressive table for sustained load; it does not itself define curve points.
PPT Limit for PMF power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default
PPT Limit for PMF APU Only power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default
STT Min Power Limit in milliwatts	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit APU in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit HS2 in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.

APTS State Index 4 (0x73)

Setting	Type	Options / Values / Range
Fan Table Index	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE Selects which fan table this APTS state uses. Point it at a more aggressive table for sustained load; it does not itself define curve points.
PPT Limit for PMF power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default
PPT Limit for PMF APU Only power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default

Setting	Type	Options / Values / Range
STT Min Power Limit in milliwatts	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit APU in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit HS2 in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.

APTS State Index 5 (0x74)

Setting	Type	Options / Values / Range
Fan Table Index	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE Selects which fan table this APTS state uses. Point it at a more aggressive table for sustained load; it does not itself define curve points.
PPT Limit for PMF power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default
PPT Limit for PMF APU Only power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default
STT Min Power Limit in milliwatts	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit APU in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit HS2 in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.

APTS State Index 6 (0x75)

Setting	Type	Options / Values / Range
Fan Table Index	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE Selects which fan table this APTS state uses. Point it at a more aggressive table for sustained load; it does not itself define curve points.
PPT Limit for PMF power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default
PPT Limit for PMF APU Only power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default

Setting	Type	Options / Values / Range
STT Min Power Limit in milliwatts	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit APU in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit HS2 in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.

APTS State Index 7 (0x76)

Setting	Type	Options / Values / Range
Fan Table Index	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE Selects which fan table this APTS state uses. Point it at a more aggressive table for sustained load; it does not itself define curve points.
PPT Limit for PMF power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default
PPT Limit for PMF APU Only power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default
STT Min Power Limit in milliwatts	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit APU in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit HS2 in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.

APTS State Index 8 (0x77)

Setting	Type	Options / Values / Range
Fan Table Index	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE Selects which fan table this APTS state uses. Point it at a more aggressive table for sustained load; it does not itself define curve points.
PPT Limit for PMF power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default
PPT Limit for PMF APU Only power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default

Setting	Type	Options / Values / Range
STT Min Power Limit in milliwatts	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit APU in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit HS2 in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.

APTS State Index 9 (0x78)

Setting	Type	Options / Values / Range
Fan Table Index	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE Selects which fan table this APTS state uses. Point it at a more aggressive table for sustained load; it does not itself define curve points.
PPT Limit for PMF power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default
PPT Limit for PMF APU Only power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default
STT Min Power Limit in milliwatts	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit APU in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit HS2 in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.

APTS State Index 10 (0x79)

Setting	Type	Options / Values / Range
Fan Table Index	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE Selects which fan table this APTS state uses. Point it at a more aggressive table for sustained load; it does not itself define curve points.
PPT Limit for PMF power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default
PPT Limit for PMF APU Only power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default

Setting	Type	Options / Values / Range
STT Min Power Limit in milliwatts	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit APU in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit HS2 in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.

APTS State Index 11 (0x7A)

Setting	Type	Options / Values / Range
Fan Table Index	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE Selects which fan table this APTS state uses. Point it at a more aggressive table for sustained load; it does not itself define curve points.
PPT Limit for PMF power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default
PPT Limit for PMF APU Only power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default
STT Min Power Limit in milliwatts	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit APU in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit HS2 in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.

APTS State Index 12 (0x7B)

Setting	Type	Options / Values / Range
Fan Table Index	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE Selects which fan table this APTS state uses. Point it at a more aggressive table for sustained load; it does not itself define curve points.
PPT Limit for PMF power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default
PPT Limit for PMF APU Only power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default

Setting	Type	Options / Values / Range
STT Min Power Limit in milliwatts	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit APU in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit HS2 in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.

APTS State Index 13 (0x7C)

Setting	Type	Options / Values / Range
Fan Table Index	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE Selects which fan table this APTS state uses. Point it at a more aggressive table for sustained load; it does not itself define curve points.
PPT Limit for PMF power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default
PPT Limit for PMF APU Only power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default
STT Min Power Limit in milliwatts	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit APU in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit HS2 in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.

APTS State Index 14 (0x7D)

Setting	Type	Options / Values / Range
Fan Table Index	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE Selects which fan table this APTS state uses. Point it at a more aggressive table for sustained load; it does not itself define curve points.
PPT Limit for PMF power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default
PPT Limit for PMF APU Only power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default

Setting	Type	Options / Values / Range
STT Min Power Limit in milliwatts	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit APU in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit HS2 in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.

APTS State Index 15 (0x7E)

Setting	Type	Options / Values / Range
Fan Table Index	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE Selects which fan table this APTS state uses. Point it at a more aggressive table for sustained load; it does not itself define curve points.
PPT Limit for PMF power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default
PPT Limit for PMF APU Only power throttler	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default
STT Min Power Limit in milliwatts	Numeric	Min 0x0 · Max 0xFFFFFFFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit APU in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.
STT Skin Temp Limit HS2 in Celsius	Numeric	Min 0x0 · Max 0xFF · Step 0x1 · Default 0x0 ■ default ■ performance TUNE STT thermal/power limit. Raising the skin-temp or min-power limit lets the APU sustain more power (more performance, hotter chassis); default protects surface temperature and safety.

DASH / ASF Management

1 forms · 8 settings

Dash Function (0x34)

Setting	Type	Options / Values / Range
Dash Function Support	Selection	Disabled [0x0] ■ default Enabled [0x1]
Disable Multi-Function	Selection	Disabled [0x0] Enabled [0x1] ■ default
Bus number Of LAN Controller	Numeric	Min 0x0 · Max 0xFF · Default 0x0 ■ default
Dev number Of LAN Controller	Numeric	Min 0x0 · Max 0x1F · Default 0x0 ■ default
Function number Of LAN Controller	Numeric	Min 0x0 · Max 0x7 · Default 0x0 ■ default
Asf Message Function Support	Selection	Disabled [0x0] ■ default Enabled [0x1]
Asf BIOS WatchDog Message Time Out Value	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■ default
Asf OS WatchDog Message Time Out Value	Numeric	Min 0x0 · Max 0xFFFF · Default 0x0 ■ default

RAIDXpert2 Configuration Utility

29 forms · 29 settings

Configuration Options (0x1)

Submenus Controller Management ■ Array Management ■ Physical Disk Management

Controller Management (0x100)

Submenus View Controller Information ■ Rescan Disks

View Controller Information (0x110)

Setting	Type	Options / Values / Range
Select Controller:	Selection	(populated at runtime)

Change Controller Properties (0x120)

Submenus Apply Changes

Convert to Legacy Disk (0x130)

Submenus OK ■ Cancel ■ Cancel

Submenus Yes ■ Yes ■ Yes ■ Yes ■ No

Convert to Legacy Disk (0x131)

Submenus OK ■ Cancel ■ Cancel

Submenus Yes ■ Yes ■ Yes ■ Yes ■ No

Initialize Disk(s) (0x140)

Submenus OK ■ Cancel ■ Cancel

Submenus Yes ■ Yes ■ Yes ■ Yes ■ No

Initialize Disk(s) (0x141)

Submenus OK ■ Cancel ■ Cancel

Submenus Yes ■ Yes ■ Yes ■ Yes ■ No

Rescan Disks (0x150)

Submenus OK

Update Controller Firmware (0x160)

Submenus Select Firmware File ■ Proceed with Firmware Download ■ Cancel Firmware Download ■ Activate Firmware ■ Cancel Firmware Activation

Setting	Type	Options / Values / Range
Firmware file extension	Selection	*.* [0x0] ■ default *.fluf [0x1] *.fw [0x2] *.bin [0x3] *.raw [0x4]

Array Management (0x200)

Submenus Create Array ■ Manage Array Properties ■ Delete Array

Create Array (0x210)

Submenus Create Array ■ Create Array ■ Select Physical Disks ■ Select Physical Disks ■ Create Array ■ Create Array ■

Submenus YES NO

Setting	Type	Options / Values / Range
Select RAID Level:	Selection	Volume [0x8] ■ default RAIDABLE [0xD] RAID 0 [0x1] RAID 1 [0x2] RAID 5 [0x3] RAID 6 [0xB] RAID 10 [0x4]
Array Size:	Numeric	Min 0x0 · Max 0xFFFFFFFFFFFFFFFF · value populated at runtime
Array Size Unit:	Selection	MB (MegaBytes) [0x2]
Select CacheTagSize:	Selection	(populated at runtime) [0x0] ■ default 64KB [0x1] 128KB [0x2] 256KB [0x3]
Select CacheTagSize:	Selection	64KB [0x1] ■ default 128KB [0x2] 256KB [0x3]
Read Cache Policy:	Selection	Read Cache [0x2] ■ default No Cache [0x1]
Write Cache Policy:	Selection	Write Back Cache [0x4] ■ default No Cache [0x3]

Select Physical Disks (0x212)

Submenus Apply Changes ■ Apply Changes ■ Apply Changes ■ Apply Changes

Submenus Apply Changes ■ Apply Changes ■

Submenus OK OK ■■

Submenus Yes No No

Setting	Type	Options / Values / Range
Select Media Type:	Selection	HDD [0x1] ■ default SSD [0x2] BOTH [0x3]

Select Physical Disks (0x211)

Submenus Apply Changes ■ Apply Changes ■ Apply Changes ■ Apply Changes

Submenus Apply Changes ■ Apply Changes ■

Submenus OK OK ■■

Submenus Yes No No

Setting	Type	Options / Values / Range
Select Media Type:	Selection	HDD [0x1] ■ default SSD [0x2] BOTH [0x3]

Select Physical Disks (0x213)

Submenus Apply Changes ■ Apply Changes ■ Apply Changes ■ Apply Changes

Submenus Apply Changes ■ Apply Changes ■

Submenus OK OK ■■

Submenus Yes No No

Select Physical Disks (0x214)

Submenus Apply Changes ■ Apply Changes ■ Apply Changes ■ Apply Changes

Submenus Apply Changes ■ Apply Changes ■

Submenus OK OK ■■

Submenus Yes No No

Create Array (0x215)

Submenus Create Array ■ Create Array ■ Select Physical Disks ■ Select Physical Disks ■ Create Array ■ Create Array ■

Submenus YES NO

Manage Array Properties (0x220)

Submenus View Associated Physical Disks Manage Dedicated Hot Spares

Setting	Type	Options / Values / Range
Select Array:	Selection	VIRT 0 01234567890123456789012345678901234567890123456789012345678901234567890 [0x1] ■ default VIRT 1 01234567890123456789012345678901234567890123456789012345678901234567890 [0x2] VIRT 2 01234567890123456789012345678901234567890123456789012345678901234567890 [0x3] VIRT 3 01234567890123456789012345678901234567890123456789012345678901234567890 [0x4] VIRT 4 01234567890123456789012345678901234567890123456789012345678901234567890 [0x5] VIRT 5 01234567890123456789012345678901234567890123456789012345678901234567890 [0x6] VIRT 6 01234567890123456789012345678901234567890123456789012345678901234567890 [0x7] VIRT 7 01234567890123456789012345678901234567890123456789012345678901234567890 [0x8] VIRT 8 01234567890123456789012345678901234567890123456789012345678901234567890 [0x9] VIRT 9 01234567890123456789012345678901234567890123456789012345678901234567890 [0xA] VIRT 10 01234567890123456789012345678901234567890123456789012345678901234567890 [0xB] VIRT 11 01234567890123456789012345678901234567890123456789012345678901234567890 [0xC] VIRT 12 01234567890123456789012345678901234567890123456789012345678901234567890 [0xD] VIRT 13 01234567890123456789012345678901234567890123456789012345678901234567890 [0xE] VIRT 14 01234567890123456789012345678901234567890123456789012345678901234567890 [0xF] VIRT 15 01234567890123456789012345678901234567890123456789012345678901234567890 [0x10] VIRT 16 01234567890123456789012345678901234567890123456789012345678901234567890 [0x11] VIRT 17 01234567890123456789012345678901234567890123456789012345678901234567890 [0x12] VIRT 18 01234567890123456789012345678901234567890123456789012345678901234567890 [0x13] VIRT 19 01234567890123456789012345678901234567890123456789012345678901234567890 [0x14] VIRT 20 01234567890123456789012345678901234567890123456789012345678901234567890 [0x15] VIRT 21 01234567890123456789012345678901234567890123456789012345678901234567890 [0x16] VIRT 22 01234567890123456789012345678901234567890123456789012345678901234567890 [0x17]

Setting	Type	Options / Values / Range
Select Array: (cont.)		VIRT 23 0123456789012345678901234567890123456789012345678901234567890 [0x18] VIRT 24 0123456789012345678901234567890123456789012345678901234567890 [0x19] VIRT 25 0123456789012345678901234567890123456789012345678901234567890 [0x1A] VIRT 26 0123456789012345678901234567890123456789012345678901234567890 [0x1B] VIRT 27 0123456789012345678901234567890123456789012345678901234567890 [0x1C] VIRT 28 0123456789012345678901234567890123456789012345678901234567890 [0x1D] VIRT 29 0123456789012345678901234567890123456789012345678901234567890 [0x1E] VIRT 30 0123456789012345678901234567890123456789012345678901234567890 [0x1F] VIRT 31 0123456789012345678901234567890123456789012345678901234567890 [0x20]

View Associated Physical Disks (0x222)

Submenus View Physical Disk Properties

Setting	Type	Options / Values / Range
Array ID:	Text	Text input · length 0x1–0x40
RAID Level:	Text	Text input · length 0x1–0x40
Array Status:	Text	Text input · length 0x1–0x40
Array Capacity:	Text	Text input · length 0x1–0x40
Cache Tag Size:	Text	Text input · length 0x1–0x40
Hidden:	Text	Text input · length 0x1–0x40
Read Cache Policy:	Text	Text input · length 0x1–0x40
Write Cache Policy:	Text	Text input · length 0x1–0x40

Manage Dedicated Hot Spares (0x224)

Submenus Add Hot Spare Disk

Submenus YES ■ NO

Manage Dedicated Hot Spares (0x225)

Submenus Add Hot Spare Disk

Submenus YES ■ NO

Delete Array (0x230)

Submenus Delete Array(s)

Warning (0x231)

Submenus YES ■ NO

View Array Properties (0x240)

Setting	Type	Options / Values / Range
Select Array:	Selection	VIRT 0 01234567890123456789012345678901234567890123456789012345678901234567890 [0x1] ■ default VIRT 1 01234567890123456789012345678901234567890123456789012345678901234567890 [0x2] VIRT 2 01234567890123456789012345678901234567890123456789012345678901234567890 [0x3] VIRT 3 01234567890123456789012345678901234567890123456789012345678901234567890 [0x4] VIRT 4 01234567890123456789012345678901234567890123456789012345678901234567890 [0x5] VIRT 5 01234567890123456789012345678901234567890123456789012345678901234567890 [0x6] VIRT 6 01234567890123456789012345678901234567890123456789012345678901234567890 [0x7] VIRT 7 01234567890123456789012345678901234567890123456789012345678901234567890 [0x8] VIRT 8 01234567890123456789012345678901234567890123456789012345678901234567890 [0x9] VIRT 9 01234567890123456789012345678901234567890123456789012345678901234567890 [0xA] VIRT 10 01234567890123456789012345678901234567890123456789012345678901234567890 [0xB] VIRT 11 01234567890123456789012345678901234567890123456789012345678901234567890 [0xC] VIRT 12 01234567890123456789012345678901234567890123456789012345678901234567890 [0xD] VIRT 13 01234567890123456789012345678901234567890123456789012345678901234567890 [0xE] VIRT 14 01234567890123456789012345678901234567890123456789012345678901234567890 [0xF] VIRT 15 01234567890123456789012345678901234567890123456789012345678901234567890 [0x10] VIRT 16 01234567890123456789012345678901234567890123456789012345678901234567890 [0x11] VIRT 17 01234567890123456789012345678901234567890123456789012345678901234567890 [0x12] VIRT 18 01234567890123456789012345678901234567890123456789012345678901234567890 [0x13] VIRT 19 01234567890123456789012345678901234567890123456789012345678901234567890 [0x14] VIRT 20 01234567890123456789012345678901234567890123456789012345678901234567890 [0x15] VIRT 21 01234567890123456789012345678901234567890123456789012345678901234567890 [0x16] VIRT 22 01234567890123456789012345678901234567890123456789012345678901234567890 [0x17]

Setting	Type	Options / Values / Range
Select Array: (cont.)		VIRT 23 01234567890123456789012345678901234567890123456789012345678901234567890 [0x18] VIRT 24 01234567890123456789012345678901234567890123456789012345678901234567890 [0x19] VIRT 25 01234567890123456789012345678901234567890123456789012345678901234567890 [0x1A] VIRT 26 01234567890123456789012345678901234567890123456789012345678901234567890 [0x1B] VIRT 27 01234567890123456789012345678901234567890123456789012345678901234567890 [0x1C] VIRT 28 01234567890123456789012345678901234567890123456789012345678901234567890 [0x1D] VIRT 29 01234567890123456789012345678901234567890123456789012345678901234567890 [0x1E] VIRT 30 01234567890123456789012345678901234567890123456789012345678901234567890 [0x1F] VIRT 31 01234567890123456789012345678901234567890123456789012345678901234567890 [0x20]
Array ID:	Text	Text input · length 0x1–0x40
RAID Level:	Text	Text input · length 0x1–0x40
Array Status:	Text	Text input · length 0x1–0x40
Array Capacity:	Text	Text input · length 0x1–0x40
Read Cache Policy:	Text	Text input · length 0x1–0x40
Write Cache Policy:	Text	Text input · length 0x1–0x40

Physical Disk Management (0x300)

Submenus [View Physical Disk Properties](#) ■ [Select Physical Disk Operations](#) ■ [View Global Hot Spares](#)

View Physical Disk Properties (0x310)

Setting	Type	Options / Values / Range
Select Physical Disk:	Selection	Physical Disk 0: XX: SATA, Size, Ready [0x1] ■ default Physical Disk 1: XX: SATA, Size, Ready [0x2] Physical Disk 2: XX: SATA, Size, Ready [0x3] Physical Disk 3: XX: SATA, Size, Ready [0x4] Physical Disk 4: XX: SATA, Size, Ready [0x5] Physical Disk 5: XX: SATA, Size, Ready [0x6] Physical Disk 6: XX: SATA, Size, Ready [0x7] Physical Disk 7: XX: SATA, Size, Ready [0x8] Physical Disk 8: XX: SATA, Size, Ready [0x9] Physical Disk 9: XX: SATA, Size, Ready [0xA] Physical Disk 10: XX: SATA, Size, Ready [0xB] Physical Disk 11: XX: SATA, Size, Ready [0xC] Physical Disk 12: XX: SATA, Size, Ready [0xD] Physical Disk 13: XX: SATA, Size, Ready [0xE] Physical Disk 14: XX: SATA, Size, Ready [0xF] Physical Disk 15: XX: SATA, Size, Ready [0x10] Physical Disk 16: XX: SATA, Size, Ready [0x11] Physical Disk 17: XX: SATA, Size, Ready [0x12] Physical Disk 18: XX: SATA, Size, Ready [0x13] Physical Disk 19: XX: SATA, Size, Ready [0x14] Physical Disk 20: XX: SATA, Size, Ready [0x15] Physical Disk 21: XX: SATA, Size, Ready [0x16] Physical Disk 22: XX: SATA, Size, Ready [0x17] Physical Disk 23: XX: SATA, Size, Ready [0x18] Physical Disk 24: XX: SATA, Size, Ready [0x19] Physical Disk 25: XX: SATA, Size, Ready [0x1A] Physical Disk 26: XX: SATA, Size, Ready [0x1B] Physical Disk 27: XX: SATA, Size, Ready [0x1C] Physical Disk 28: XX: SATA, Size, Ready [0x1D] Physical Disk 29: XX: SATA, Size, Ready [0x1E] Physical Disk 30: XX: SATA, Size, Ready [0x1F] Physical Disk 31: XX: SATA, Size, Ready [0x20]

Select Physical Disk Operations (0x320)

Submenus Initialize Disk ■ Assign Global Hot Spare ■ Assign Global Hot Spare ■ Unassign Hot Spare ■

Submenus Yes No 111113 GTRPRPI1001C · AMI Aptio V · AMD Strix Halo

Submenus Yes ■ No 112113 GTRPRPI1001C · AMI Aptio V · AMD Strix Halo Form / PagePurposeProvider Module NetworkStackSetu
 Network Stack ConfigurationIPv4 / IPv6 network device policy pScreen Ip4Dxe IPv4 Network ConfigurationStatic IP, netmask, gateway, DNS
 (per NIC) Ip6Dxe IPv6 Network ConfigurationIPv6 interface policy and addressing (per NIC) VLAN ConfigurationVLAN ID and priority (per
 NIC)VlanConfigDxe HTTP Boot ConfigurationUEFI HTTP/HTTPS boot URI and parametersHttpBootDxe TLS Auth ConfigurationServer CA /
 client certificate enrollmentTlsAuthConfigDxe PXE / Network BootCRB PXE UEFI boot driver optionsCrbPxeUefiDriver 113113

Setting	Type	Options / Values / Range
Select Physical Disk:	Selection	Physical Disk 0: XX: SATA, Size, Ready [0x1] ■ default Physical Disk 1: XX: SATA, Size, Ready [0x2] Physical Disk 2: XX: SATA, Size, Ready [0x3] Physical Disk 3: XX: SATA, Size, Ready [0x4] Physical Disk 4: XX: SATA, Size, Ready [0x5] Physical Disk 5: XX: SATA, Size, Ready [0x6] Physical Disk 6: XX: SATA, Size, Ready [0x7] Physical Disk 7: XX: SATA, Size, Ready [0x8] Physical Disk 8: XX: SATA, Size, Ready [0x9] Physical Disk 9: XX: SATA, Size, Ready [0xA] Physical Disk 10: XX: SATA, Size, Ready [0xB] Physical Disk 11: XX: SATA, Size, Ready [0xC] Physical Disk 12: XX: SATA, Size, Ready [0xD] Physical Disk 13: XX: SATA, Size, Ready [0xE] Physical Disk 14: XX: SATA, Size, Ready [0xF] Physical Disk 15: XX: SATA, Size, Ready [0x10] Physical Disk 16: XX: SATA, Size, Ready [0x11] Physical Disk 17: XX: SATA, Size, Ready [0x12] Physical Disk 18: XX: SATA, Size, Ready [0x13] Physical Disk 19: XX: SATA, Size, Ready [0x14] Physical Disk 20: XX: SATA, Size, Ready [0x15] Physical Disk 21: XX: SATA, Size, Ready [0x16] Physical Disk 22: XX: SATA, Size, Ready [0x17] Physical Disk 23: XX: SATA, Size, Ready [0x18] Physical Disk 24: XX: SATA, Size, Ready [0x19] Physical Disk 25: XX: SATA, Size, Ready [0x1A] Physical Disk 26: XX: SATA, Size, Ready [0x1B] Physical Disk 27: XX: SATA, Size, Ready [0x1C] Physical Disk 28: XX: SATA, Size, Ready [0x1D] Physical Disk 29: XX: SATA, Size, Ready [0x1E] Physical Disk 30: XX: SATA, Size, Ready [0x1F] Physical Disk 31: XX: SATA, Size, Ready [0x20]

Select Physical Disk Operations (0x321)

Submenus Initialize Disk ■ Assign Global Hot Spare ■ Assign Global Hot Spare ■ Unassign Hot Spare ■

Submenus Yes No 111113 GTRPRPI1001C · AMI Aptio V · AMD Strix Halo

Submenus Yes ■ No 112113 GTRPRPI1001C · AMI Aptio V · AMD Strix Halo Form / PagePurposeProvider Module NetworkStackSetu Network Stack ConfigurationIPv4 / IPv6 network device policy pScreen Ip4Dxe IPv4 Network ConfigurationStatic IP, netmask, gateway, DNS (per NIC) Ip6Dxe IPv6 Network ConfigurationIPv6 interface policy and addressing (per NIC) VLAN ConfigurationVLAN ID and priority (per NIC)VlanConfigDxe HTTP Boot ConfigurationUEFI HTTP/HTTPS boot URI and parametersHttpBootDxe TLS Auth ConfigurationServer CA / client certificate enrollmentTlsAuthConfigDxe PXE / Network BootCRB PXE UEFI boot driver optionsCrbPxeUefiDriver 113113

Select Physical Disk Operations (0x322)

Submenus Initialize Disk ■ Assign Global Hot Spare ■ Assign Global Hot Spare ■ Unassign Hot Spare ■

Submenus Yes No 111113 GTRPRPI1001C · AMI Aptio V · AMD Strix Halo

Submenus Yes ■ No 112113 GTRPRPI1001C · AMI Aptio V · AMD Strix Halo Form / PagePurposeProvider Module NetworkStackSetu Network Stack ConfigurationIPv4 / IPv6 network device policy pScreen Ip4Dxe IPv4 Network ConfigurationStatic IP, netmask, gateway, DNS (per NIC) Ip6Dxe IPv6 Network ConfigurationIPv6 interface policy and addressing (per NIC) VLAN ConfigurationVLAN ID and priority (per NIC)VlanConfigDxe HTTP Boot ConfigurationUEFI HTTP/HTTPS boot URI and parametersHttpBootDxe TLS Auth ConfigurationServer CA / client certificate enrollmentTlsAuthConfigDxe PXE / Network BootCRB PXE UEFI boot driver optionsCrbPxeUefiDriver 113113

Appendix A — Dynamic Driver Forms (Excluded)

The forms below exist in the firmware but are contributed by generic UEFI network and boot drivers rather than by this board's BIOS. They surface only when the relevant controller or driver binds at runtime, and their contents are standardised by the UEFI driver model. They are listed for completeness and are not enumerated option-by-option, and they are not counted in the settings totals.

Form / Page	Purpose	Provider module
Network Stack Configuration	IPv4 / IPv6 network device policy	NetworkStackSetupScreen
IPv4 Network Configuration	Static IP, netmask, gateway, DNS (per NIC)	Ip4Dxe
IPv6 Network Configuration	IPv6 interface policy and addressing (per NIC)	Ip6Dxe
VLAN Configuration	VLAN ID and priority (per NIC)	VlanConfigDxe
HTTP Boot Configuration	UEFI HTTP/HTTPS boot URI and parameters	HttpBootDxe
TLS Auth Configuration	Server CA / client certificate enrolment	TlsAuthConfigDxe
PXE / Network Boot	CRB PXE UEFI boot driver options	CrbPxeUefiDriver